CHAPTER 04

Analyzing Organic Reactions

Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: da2bc9fa814ef298f105232378af42f8b11d4df69303deef3e612d2153bf8488

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

Coverage checklist

ch04:exercise:pka-equilibrium | pka-equilibrium:stage-1, pka-equilibrium:stage-2, pka-equilibrium:stage-3

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ch04:exercise:carbon-redox | carbon-redox:stage-1, carbon-redox:stage-2, carbon-redox:stage-3

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ch04:molecule:aceticacid | aceticacid

ch04:molecule:acetate | acetate

ch04:molecule:acetaldehyde | acetaldehyde

ch04:molecule:bromoethane | bromoethane

ch04:molecule:tbutylbromide | tbutylbromide

ch04:molecule:ethanol | ethanol

ch04:molecule:acetone | acetone

ch04:molecule:ketoester | ketoester

ch04:curriculum | scope-1, scope-2, scope-3, scope-4, scope-5

Curriculum scope

scope-1: 4.1 Acids and Bases

scope-2: 4.2 Nucleophiles, Electrophiles, and Leaving Groups

scope-3: 4.3 Oxidation–Reduction Reactions

scope-4: 4.4 Chemoselectivity

scope-5: 4.5 Steps to Problem Solving

Objective: Estimate acid–base equilibria

Objective: Distinguish SN1 and SN2

Objective: Track oxidation and chemoselectivity

Supplied Kaplan curriculum: Chapter 4; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter04b.xhtml

SN1 stereochemical qualifications: OpenStax 11.4 | https://openstax.org/books/organic-chemistry/pages/11-4-the-sn1-reaction

OpenStax: SN2 nucleophile and solvent effects | https://openstax.org/books/organic-chemistry/pages/11-3-characteristics-of-the-sn2-reaction

OpenStax: aldehyde oxidation conditions | https://openstax.org/books/organic-chemistry/pages/19-3-oxidation-of-aldehydes-and-ketones

OpenStax: amide reduction to amines | https://openstax.org/books/organic-chemistry/pages/21-7-chemistry-of-amides

OpenStax: acyl substitution and conditional reactivity order | https://openstax.org/books/organic-chemistry/pages/21-2-nucleophilic-acyl-substitution-reactions

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

acids | Acid–base reasoning

Equilibrium favors the weaker acid/base pair

acids:rule | Core Rules

Brønsted acids donate H+ and bases accept it. Lewis acids accept an electron pair; Lewis bases donate one. In a coordinate covalent bond, both shared electrons originate from the donor; the resulting bond is still covalent. For HA + B ⇌ A− + BH+, K ≈ 10^[pKa(BH+) − pKa(HA)] when the pKa values use comparable conditions. Amphoteric species can act as acids or bases, depending on the partner. Water can accept H+ to form H3O+ or donate H+ to form OH−; bicarbonate can likewise gain or lose a proton.

Visual card: Acid–Base Strength (pKa), Conjugate Base Stability (ARIO) & Equilibrium Direction

Chemistry diagram: Illustrative acid/conjugate-base pair. Protonation depends on pH; the panels are not separate resonance contributors.

$pKa = -\log_{10}(Ka)$ | $pH = pKa + \log_{10}([A^-]/[HA])$ | Equilibrium Favors HIGHER pKa (Weaker Acid)

↓ pKa = ↑ Ka = Stronger Acid

ARIO: Atom > Resonance > Induction > Orbital

pH > pKa → A^- Predominates

pH < pKa | Acidic buffer region | Protonated form (HA / RNH_3^+) predominates

pH = pKa | Half-equivalence point | [HA] = $[A^-]$ (50:50 max buffer capacity)

pH > pKa | Basic buffer region | Deprotonated conjugate base (A^- / RNH_2) wins

ARIO Priority Checklist — — 1. Atom (Row: EN C<N<O<F; Column: Size F<Cl<Br<I) → 2. Resonance → 3. Induction (EDGs vs EWGs) → 4. Orbital ($sp > sp^2 > sp^3$).

Predicting Equilibrium Direction — — Compare the two acids (HA on left vs HB^+ on right): equilibrium ALWAYS shifts toward the weaker acid (the side with the HIGHER pKa).

acids:cue | Memory Cues

Lower pKa means stronger acid. Stabilize the conjugate base to increase acidity. Across a comparable row, electronegativity can stabilize negative charge; down a group, weaker H–X bonds and charge dispersion can dominate. Compare like structures and conditions, rather than applying a single trend everywhere.

Visual card: ARIO Conjugate Base Stability & pH vs. pKa Speciation

How are you evaluating acid strength or protonation state?

Comparing Two Acids Without pKa Values — Deprotonate Both → Apply ARIO to Conjugate Base A^- — More stable conjugate base = stronger acid! Across a row electronegativity wins (OH > NH_2); down a column size wins (I^- > F^-, so HI > HF).

Comparing pH to pKa in Solution — pH < pKa → Protonated (HA); pH > pKa → Deprotonated (A^-) — If pH is 2 units above pKa, the ratio $[A^-]/[HA]$ is 10^2 = 100:1 (99% deprotonated).

Inductive Effect (CF_3COOH vs CH_3COOH) — EWGs Pull Electron Density → ↓ pKa (Stronger Acid) — Inductive withdrawal falls off rapidly with distance (α-Cl > β-Cl > γ-Cl) and follows electronegativity (F > Cl > Br > I).

acids:trap | Common Mistakes / MCAT Traps

Basicity is an equilibrium property; nucleophilicity is a reaction-rate property. Do not use a single periodic trend across different solvents or unrelated functional groups.

Visual card: Thinking HF Is Stronger Than HI or That Electron-Withdrawing Groups Raise pKa

Common Trap — — Saying HF is stronger than HI because F is more electronegative, or saying EWGs "increase pKa".

Correct Reasoning — — Down a periodic column, atomic radius dominates (I^- spreads negative charge over a huge volume, so HI pKa $-10 \ll$ HF pKa 3.2); stronger acids have LOWER pKa!

acids:app | MCAT Applications

A carboxylic acid with pKa 4.8 is mostly carboxylate at pH 7.4; that changes water solubility and membrane partitioning.

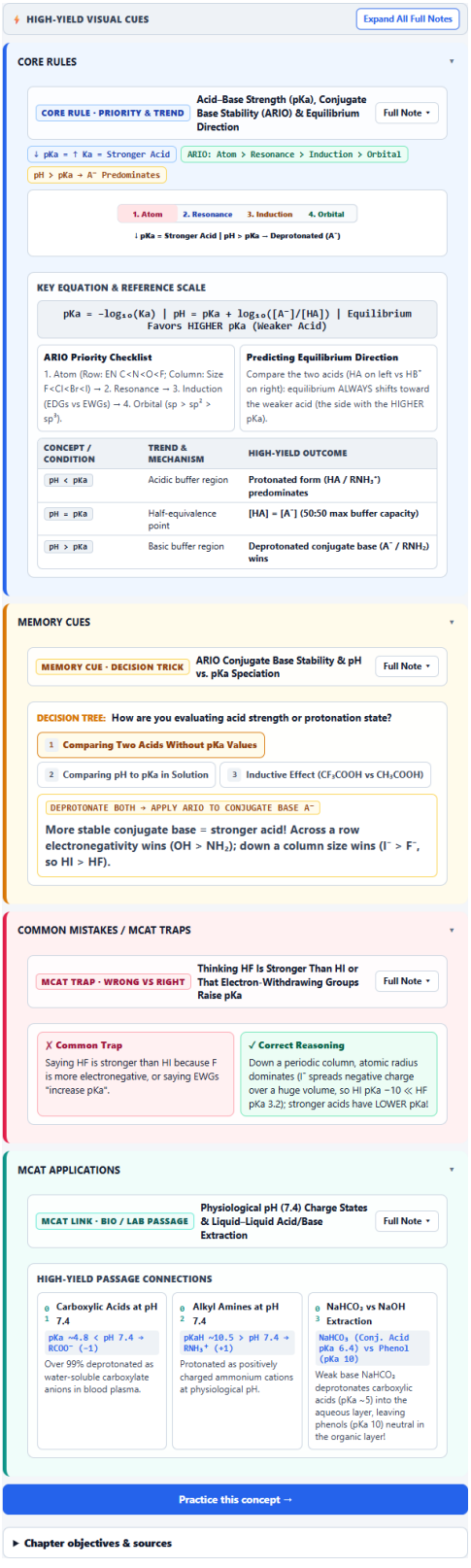
Visual card: Physiological pH (7.4) Charge States & Liquid–Liquid Acid/Base Extraction

Carboxylic Acids at pH 7.4 — pKa ~4.8 < pH 7.4 → $RCOO^-$ (−1) — Over 99% deprotonated as water-soluble carboxylate anions in blood plasma.

Alkyl Amines at pH 7.4 — pKaH ~10.5 > pH 7.4 → RNH_3^+ (+1) — Protonated as positively charged ammonium cations at physiological pH.

$NaHCO_3$ vs NaOH Extraction — $NaHCO_3$ (Conj. Acid pKa 6.4) vs Phenol (pKa 10) — Weak base $NaHCO_3$ deprotonates carboxylic acids (pKa ~5) into the aqueous layer, leaving phenols (pKa 10) neutral in the organic layer!

Screenshot: acids-visuals



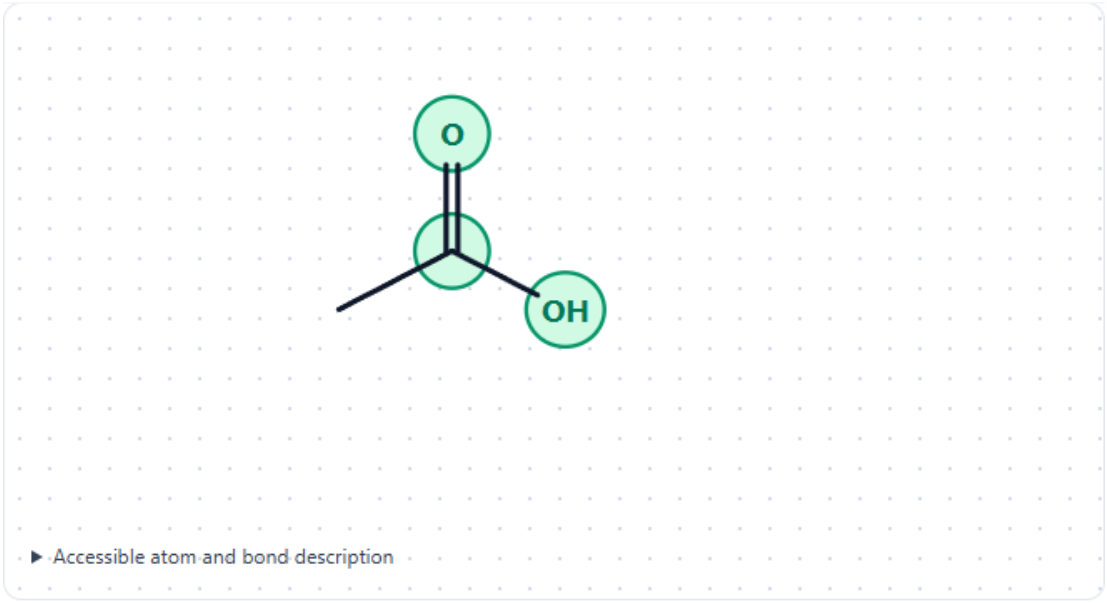
acids:step-1 | 1. Locate the acidic group

Removing the carboxyl O–H proton gives a carboxylate whose negative charge is shared across two oxygens.

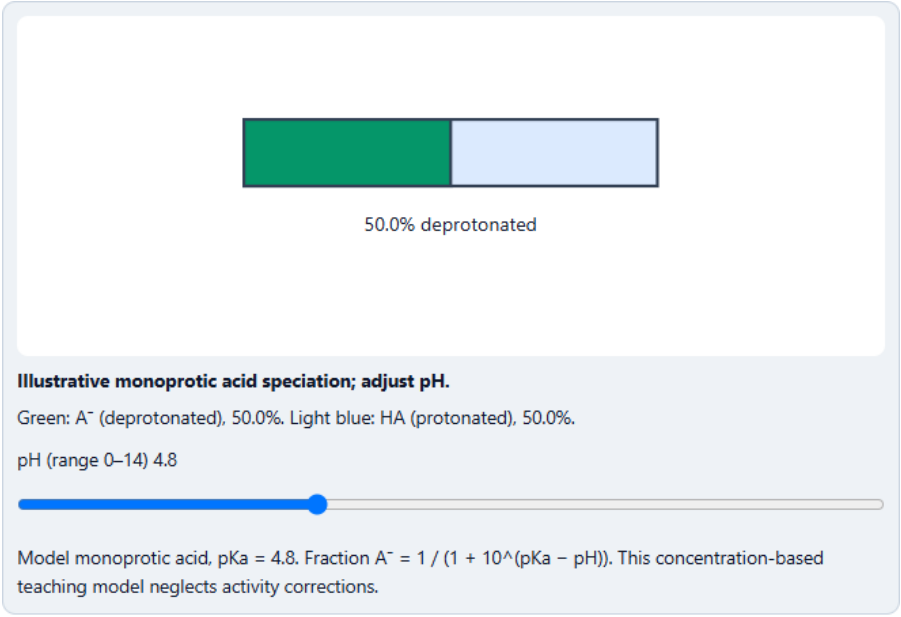
Displayed molecule ID: aceticacid

{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

Screenshot: acids-step-1



Screenshot: acids-step-1-diagram



{"diagramSpecification": {"type": "acid", "pKa": 4.8, "caption": "Illustrative monoprotic acid speciation; adjust pH."}}

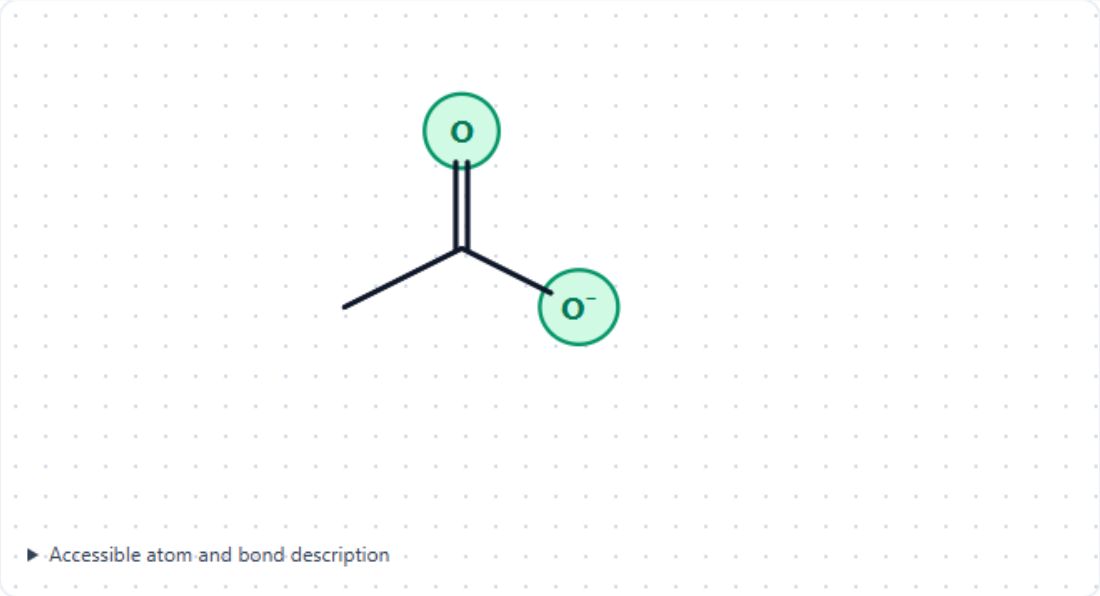
acids:step-2 | 2. Compare conjugate bases

An alcohol-derived alkoxide lacks the equivalent two-oxygen resonance of a carboxylate. This is a major reason simple carboxylic acids are more acidic than alcohols.

Displayed molecule ID: acetate

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Screenshot: acids-step-2



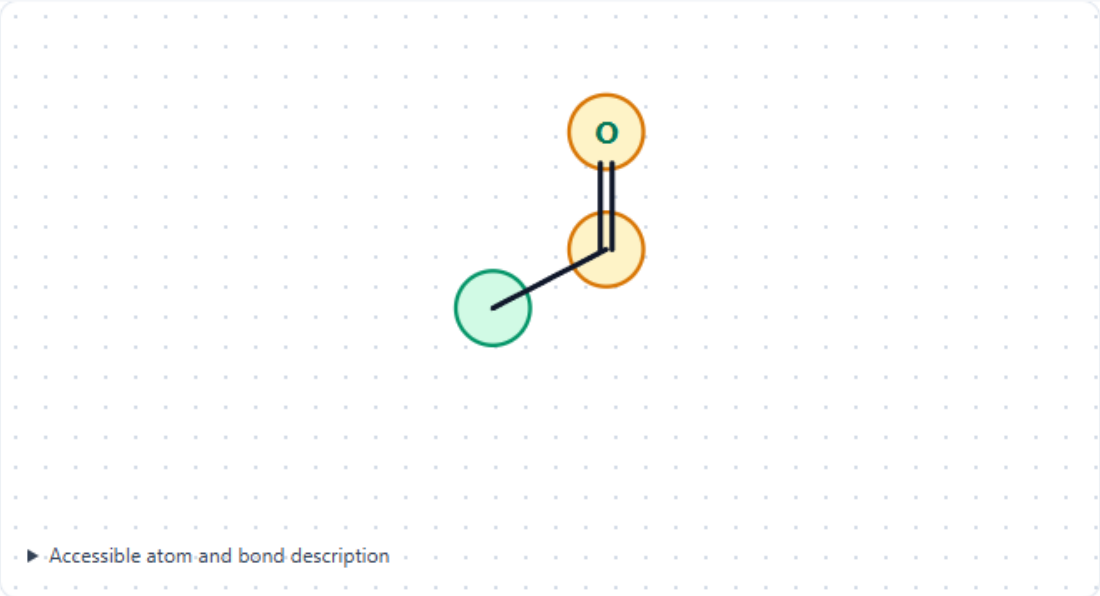
acids:step-3 | 3. Find alpha hydrogens

In ethanal, the methyl carbon is alpha to the carbonyl. Removing one of its hydrogens with a suitable base can form a resonance-stabilized enolate. Distinguish the acidic alpha C–H sites from the electrophilic carbonyl carbon; an enolate can act as a nucleophile. These hydrogens are implicit in the drawing.

Displayed molecule ID: acetaldehyde

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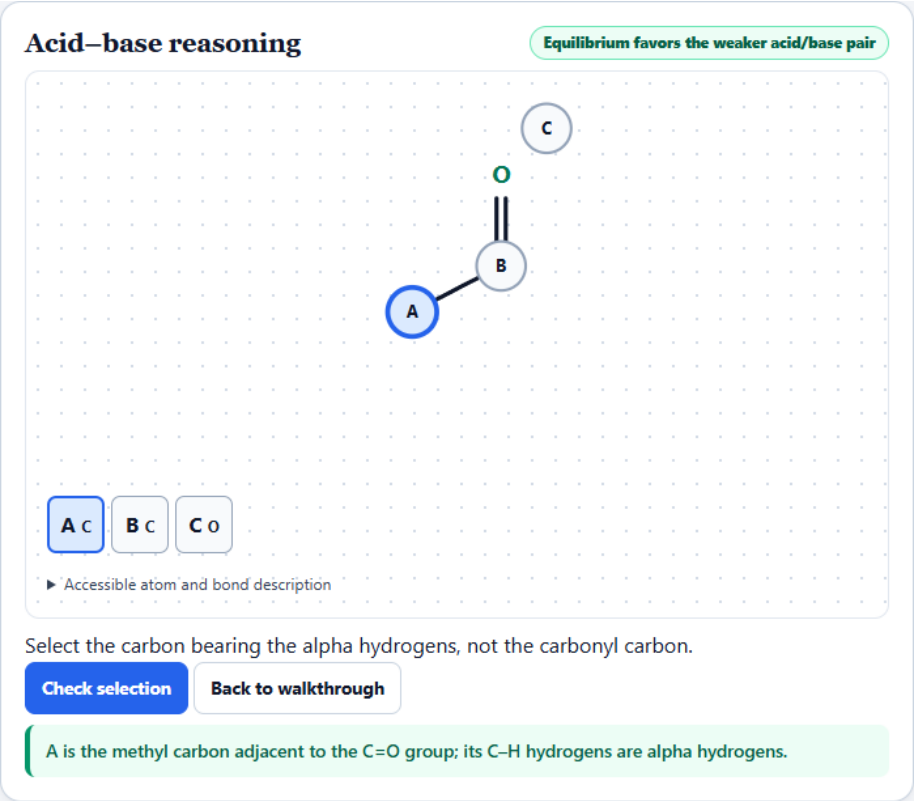
Screenshot: acids-step-3



Embedded challenge: Select the carbon bearing the alpha hydrogens, not the carbonyl carbon.

{"prompt": "Select the carbon bearing the alpha hydrogens, not the carbonyl carbon.", "answer": [0], "explanation": "A is the methyl carbon adjacent to the C=O group; its C–H hydrogens are alpha hydrogens."}

Screenshot: acids-step-3-challenge



partners | Nucleophiles, electrophiles, and leaving groups

Electron pairs move from donor to acceptor

partners:rule | Core Rules

A nucleophile supplies an electron pair to an electrophile. A leaving group takes the bonding pair when it departs. Stable weak bases usually make better leaving groups. Protonation can turn poor leaving group OH− into neutral water.

Visual card: Nucleophiles, Electrophiles, Leaving Group Ability & Solvent Effects

Chemistry diagram: Follow electron donation and the leaving-group bond in this illustrative graph.

Nucleophile = Lewis Base (e^- Donor)

Electrophile = Lewis Acid (e^- Acceptor)

Good LG = Weak Base (Low pKaH)

Protic Solvent (ROH, H_2O) | H-bonds cage small anions | Nucleophilicity: $I^- > Br^- > Cl^- > F^-$

Polar Aprotic (DMSO, DMF) | Bare unsolvated anions! | Nucleophilicity: $F^- > Cl^- > Br^- > I^-$

Good Leaving Group | Conjugate base of strong acid | TsO^-, I^-, Br^-, Cl^-, H_2O (never HO^- in SN1/SN2)

Nucleophilicity vs. Basicity — — Basicity is thermodynamic affinity for H^+; nucleophilicity is kinetic attack rate at carbon (sensitive to steric bulk and solvent).

Protic vs. Polar Aprotic Solvents — — Protic solvents H-bond tightly to small F^- ($I^- > Br^- > Cl^- > F^-$); polar aprotic solvents leave F^- bare and reactive ($F^- > Cl^- > Br^- > I^-$).

Leaving Group Rule (Weak Base = Good LG) — — The best leaving groups are stable weak bases whose conjugate acids have pKa < 0 (TsO^-, MsO^-, I^-, Br^-, Cl^-, H_2O).

partners:cue | Memory Cues

Find electron-rich donor → electron-poor acceptor → possible leaving group. Curved arrows start at electrons, not positive charges. Compare nucleophilicity using charge, reacting-atom identity, steric access, and solvent. For a similar reacting atom, an anion is often more nucleophilic than its neutral conjugate acid. Across comparable atoms in a row, lower electronegativity often favors donation. Bulky donors are hindered in SN2. Protic solvation and polarizability can reverse simple basicity-based trends down a group; polar aprotic conditions often favor anionic SN2 nucleophiles.

Visual card: Strong Nucleophile vs. Bulky Base & Leaving Group Activation

What reagent or leaving group role are you diagnosing?

Small Charged Anion (CN^-, RS^-, I^-, N_3^-, RO^-) — Strong Nucleophile (Attacks Unhindered C) — High polarizability or localized negative charge with minimal steric bulk favors substitution attack at δ+ carbon.

Sterically Hindered Base (t-BuO^-, LDA, DBU) — Strong Base Only → Forces E2 Elimination! — Too bulky to reach a backside carbon σ* orbital; plucks an exposed β-proton instead.

Poor Leaving Group (–OH, –OR, $–NH_2$) — Protonate (H^+ → $–OH_2^+$) or Tosylate (TsCl → –OTs) — Strong bases (HO^-, RO^-, H_2N^-) never leave directly in SN1/SN2; convert –OH into H_2O or –OTs first!

partners:trap | Common Mistakes / MCAT Traps

A carbonyl oxygen is electron rich while its carbon is electron poor. A positive charge is not required for electrophilicity. For comparable neutral acyl derivatives in nucleophilic acyl substitution, anhydrides are generally more reactive than esters, and esters more reactive than amides. A free carboxylic acid can instead protonate a basic nucleophile; protonation, activation and the leaving group prevent a universal acid/ester ranking.

Visual card: Kicking Out Hydroxide (HO^-) Directly in SN1/SN2 or Reversing Curved Arrows

Common Trap — — Drawing ROH + Br^- → RBr + HO^- directly, or drawing a curved arrow from the electrophile C^+ to the nucleophile.

Correct Reasoning — — Curved arrows ALWAYS start at an electron source (lone pair or π bond) and point to an electron sink; protonate –OH to $–OH_2^+$ before displacement.

partners:app | MCAT Applications

Enzyme active sites position nucleophiles and electrophiles while stabilizing charge; the same donor–acceptor logic applies.

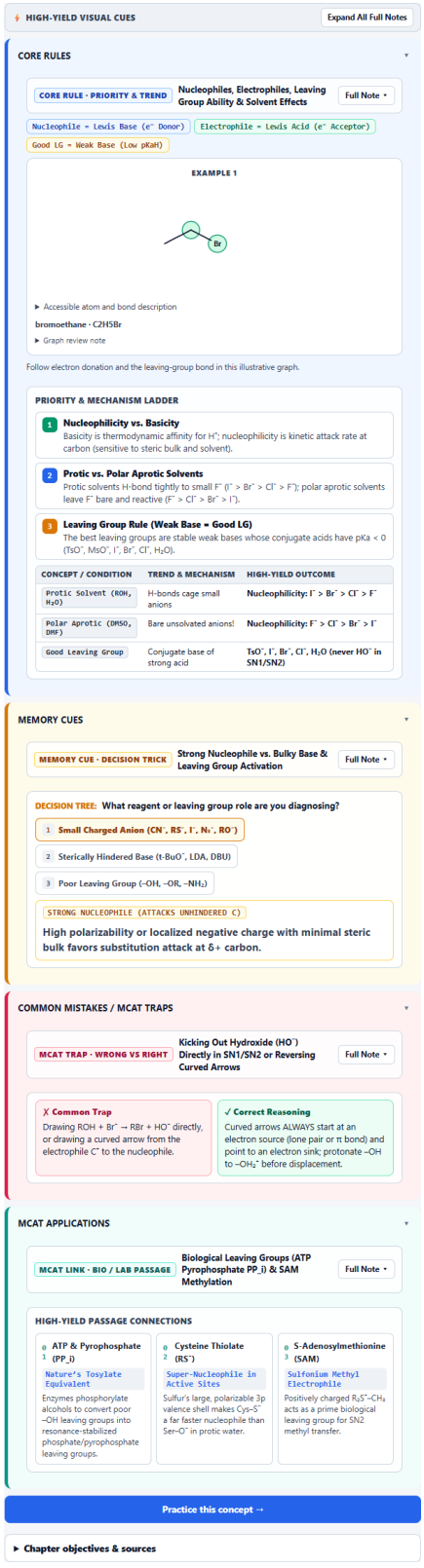
Visual card: Biological Leaving Groups (ATP Pyrophosphate PP_i) & SAM Methylation

ATP & Pyrophosphate (PP_i) — Nature's Tosylate Equivalent — Enzymes phosphorylate alcohols to convert poor –OH leaving groups into resonance-stabilized phosphate/pyrophosphate leaving groups.

Cysteine Thiolate (RS^-) — Super-Nucleophile in Active Sites — Sulfur's large, polarizable 3p valence shell makes Cys–S^- a far faster nucleophile than Ser–O^- in protic water.

S-Adenosylmethionine (SAM) — Sulfonium Methyl Electrophile — Positively charged $R_2S^+–CH_3$ acts as a prime biological leaving group for SN2 methyl transfer.

Screenshot: partners-visuals



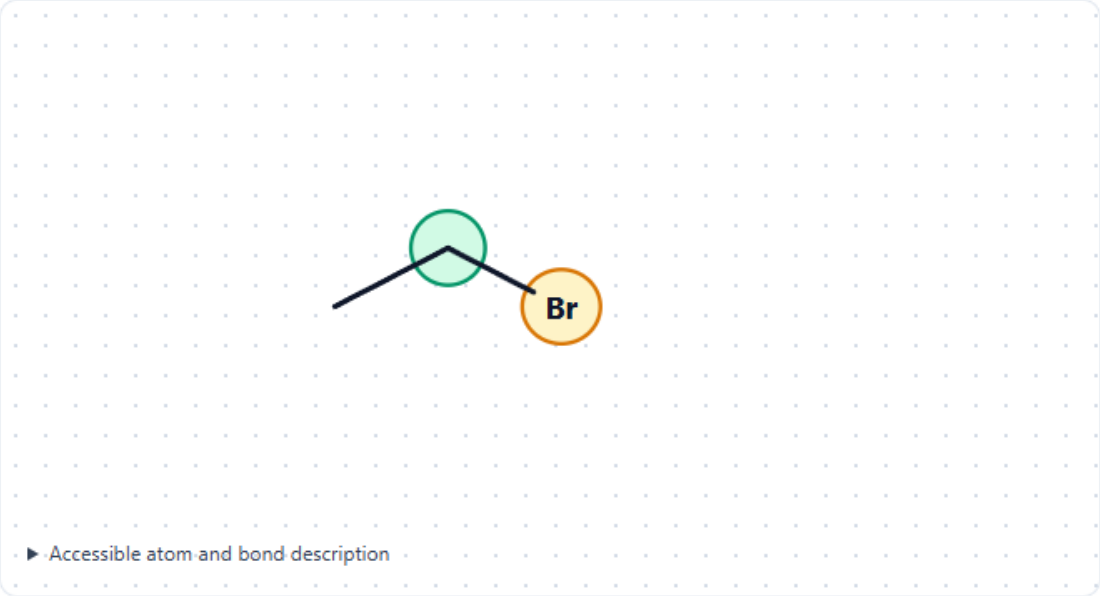
partners:step-1 | 1. Polarize the bond

Bromine withdraws electron density from the attached carbon. The C bonded to Br is the electrophilic site for substitution.

Displayed molecule ID: bromoethane

{"highlights": {"prin": [1], "sub": [2]}, "revealName": false}

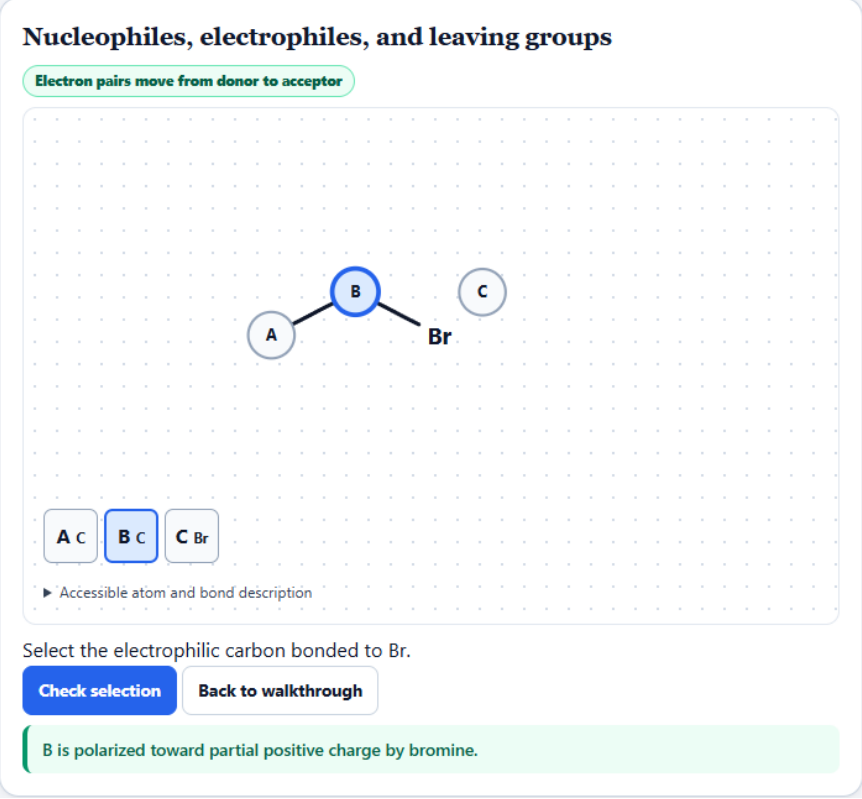
Screenshot: partners-step-1



Embedded challenge: Select the electrophilic carbon bonded to Br.

{"prompt": "Select the electrophilic carbon bonded to Br.", "answer": [1], "explanation": "B is polarized toward partial positive charge by bromine."}

Screenshot: partners-step-1-challenge



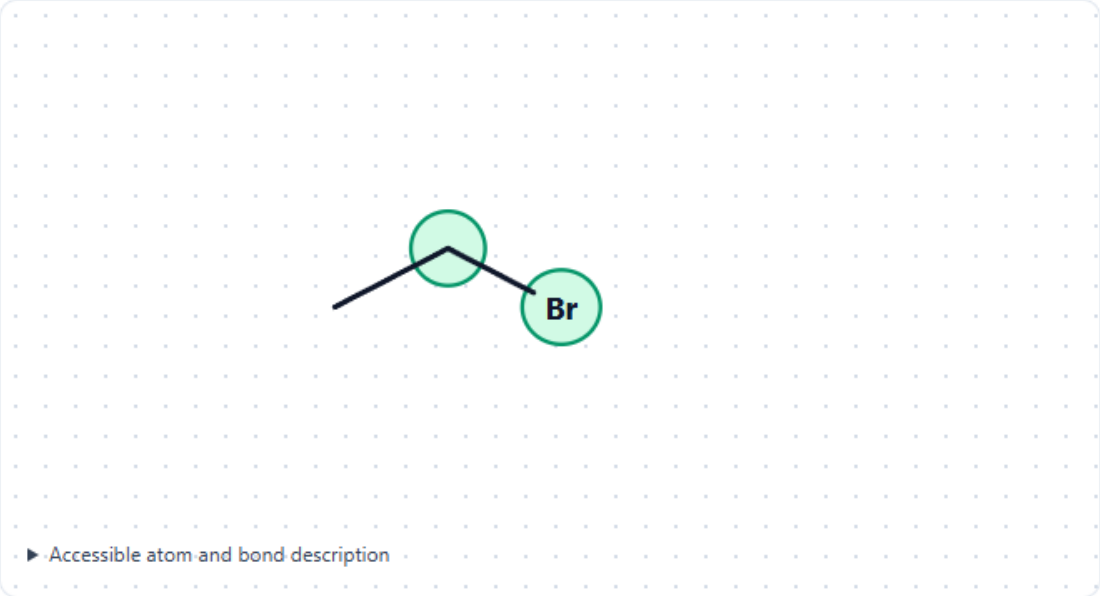
partners:step-2 | 2. Follow bond cleavage

In substitution, a nucleophile forms a bond to the electrophilic carbon while the C–Br bond ultimately breaks with its electron pair going to Br. Conditions determine whether this is concerted or stepwise.

Displayed molecule ID: bromoethane

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: partners-step-2



substitution | SN1 vs. SN2

Mechanism depends on substrate and conditions

substitution:rule | Core Rules

SN2 is concerted backside attack: rate = k[substrate][nucleophile], with geometric inversion. Methyl and unhindered primary centers are favorable. SN1 ionizes to a carbocation: rate = k[substrate]; tertiary, allylic, and benzylic stabilization can help.

Visual card: SN2 (Concerted Backside Inversion) vs. SN1 (Stepwise Carbocation Racemization)

Chemistry diagram: Compare steric access in tertiary and primary substrates; mechanism also depends on conditions.

SN2 Rate = k [Substrate][Nucleophile] (Bimolecular) | SN1 Rate = k [Substrate] (Unimolecular)

SN2: 1 Step · Rate=k[RX][Nu] · Inversion

SN1: 2 Steps · Rate=k[RX] · Racemization

3° Carbocation > 2° > 1°

SN2 (Concerted) | Methyl > 1° > 2° (No 3°!) | Strong Nu^- + Polar Aprotic → 100% Inversion

SN1 (Carbocation) | 3° > 2° (Resonance: Allylic/Benzylic) | Weak Nu (ROH/H_2O) + Polar Protic → Racemic

SN2 Stereochemistry & Substrate — — 1-step backside attack into σ* LUMO inverts the stereocenter umbrella (Methyl > 1° > 2°; 3° is blocked by steric hindrance).

SN1 Stereochemistry & Rearrangement — — Rate-determining LG loss forms a planar sp^2 carbocation (3°/allylic/benzylic > 2°); attacked from both faces (racemization) and can rearrange!

substitution:cue | Memory Cues

SN2: two species in the rate law, one concerted event. SN1: slow ionization, then attack.

Visual card: SN1 vs. SN2 4-Point Diagnostic Matrix

Check the substrate, nucleophile, and solvent:

Methyl or 1° Substrate + Strong Nu^- (CN^-, I^-, RS^-) — SN2 Mechanism (Concerted Backside Attack) — Doubling [Nu^-] doubles the rate; a chiral 1° deuterated or 2° substrate undergoes 100% configuration inversion.

3° Substrate + Weak/Neutral Solvolysis (H_2O, MeOH) — SN1 Mechanism (Planar Carbocation Intermediate) — Rate depends ONLY on [Substrate]; can partially racemize; ion pairing can bias the faces and shows two humps on the reaction coordinate diagram.

2° Substrate adjacent to 3° or Quaternary Carbon in SN1 — Watch for 1,2-Hydride or 1,2-Methyl Shift! — Carbocation rearrangement is possible when an accessible shift produces a sufficiently favored intermediate; assess the actual structure.

substitution:trap | Common Mistakes / MCAT Traps

SN1 can rearrange and often racemizes a stereocenter, but ion pairing can prevent exactly 50:50 products. SN2 inversion does not guarantee an R↔S label change if CIP priorities change. Strong bases and heat can make elimination compete.

Visual card: Running SN2 on a Tertiary Halide or Doubling SN1 Rate with Nucleophile Concentration

Common Trap — — Drawing SN2 backside attack on tert-butyl bromide or saying doubling [Nu] doubles the rate of an SN1 reaction.

Correct Reasoning — — 3° substrates NEVER undergo SN2 (strong base/nucleophiles give E2 elimination instead), and nucleophiles do not appear in the SN1 rate law!

substitution:app | MCAT Applications

Use a rate change to distinguish mechanisms: doubling nucleophile concentration doubles an ideal SN2 rate but does not alter an ideal SN1 rate.

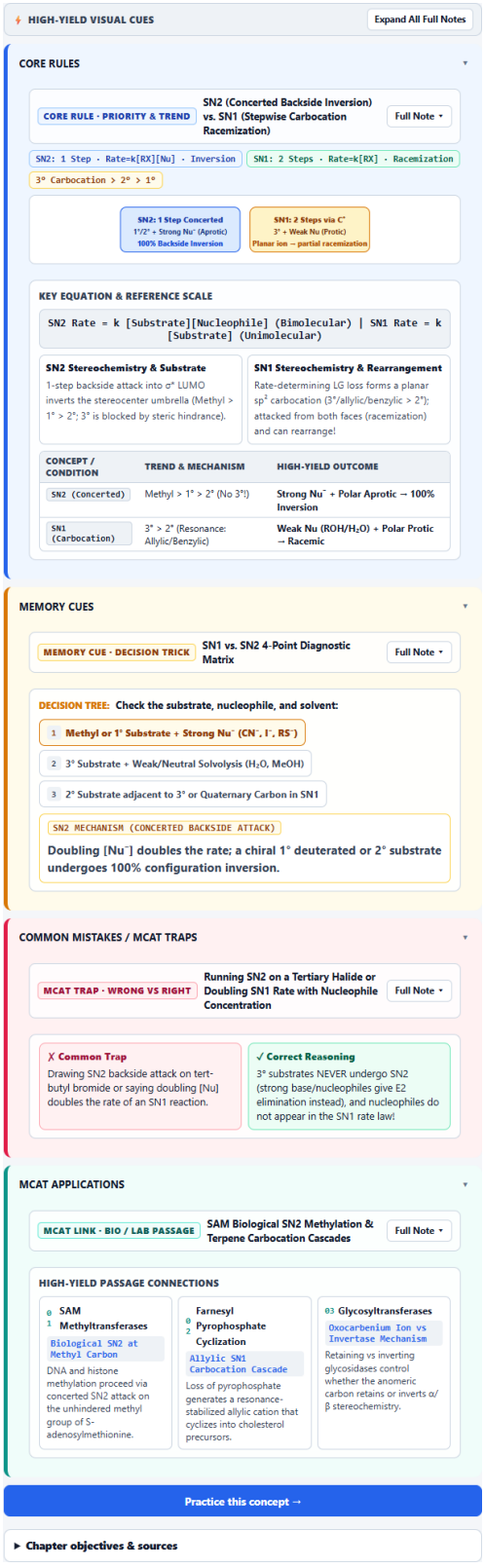
Visual card: SAM Biological SN2 Methylation & Terpene Carbocation Cascades

SAM Methyltransferases — Biological SN2 at Methyl Carbon — DNA and histone methylation proceed via concerted SN2 attack on the unhindered methyl group of S-adenosylmethionine.

Farnesyl Pyrophosphate Cyclization — Allylic SN1 Carbocation Cascade — Loss of pyrophosphate generates a resonance-stabilized allylic cation that cyclizes into cholesterol precursors.

Glycosyltransferases — Oxocarbenium Ion vs Invertase Mechanism — Retaining vs inverting glycosidases control whether the anomeric carbon retains or inverts α/β stereochemistry.

Screenshot: substitution-visuals



substitution:step-1 | 1. Inspect crowding

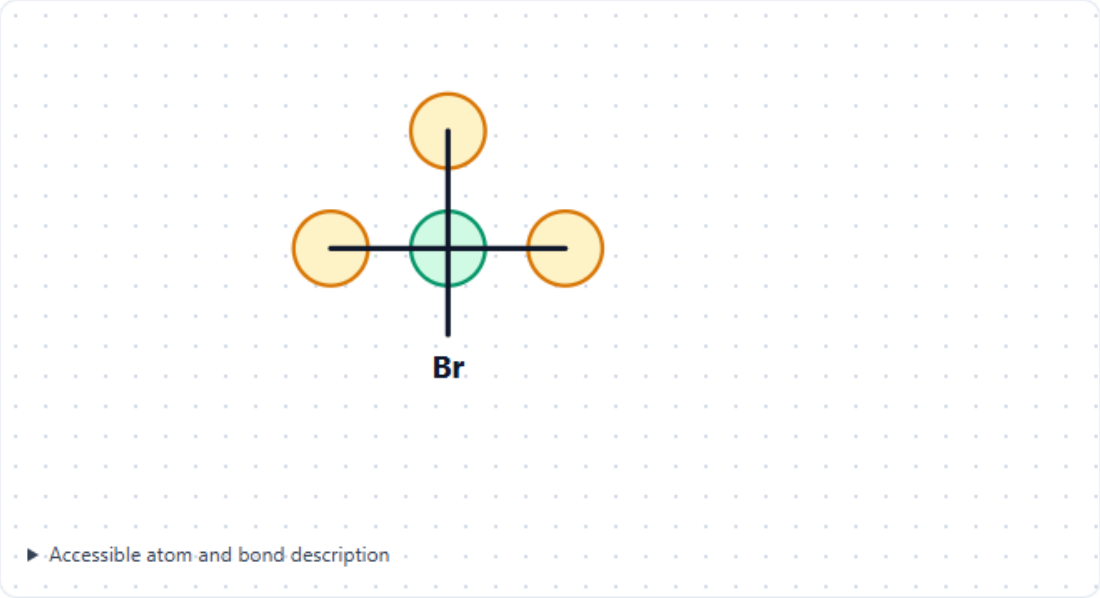
The carbon bonded to Br is tertiary: its three carbon neighbors are highlighted in orange. The cross is a 2D connectivity drawing, not a claim of 90° bond angles. Backside SN2 attack is strongly hindered; a suitable ionizing solvent can permit

formation of a tertiary carbocation.

Displayed molecule ID: tbutylbromide

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Screenshot: substitution-step-1



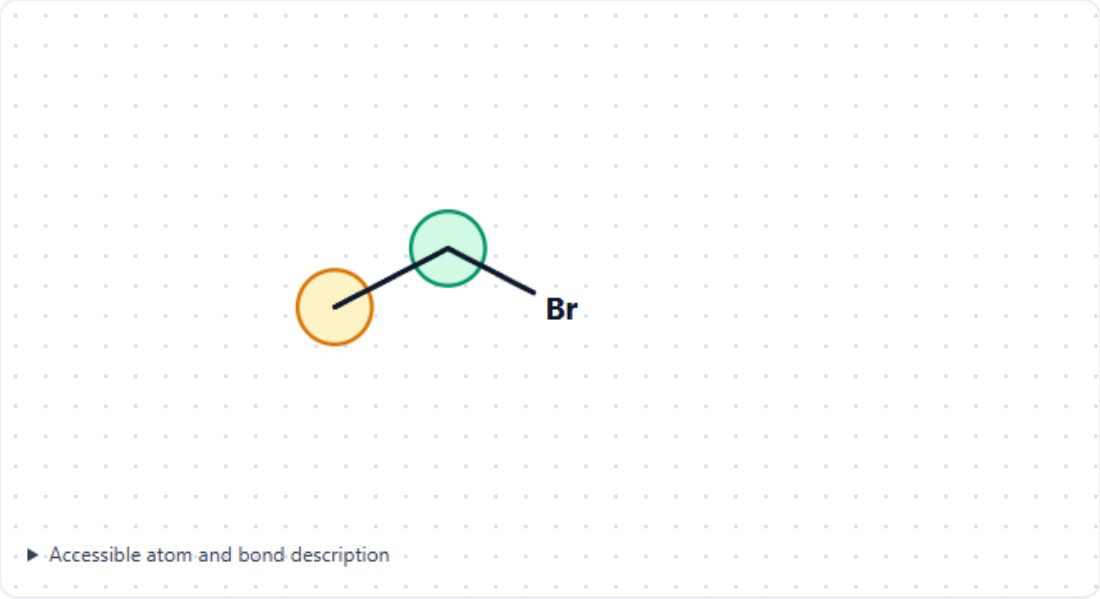
substitution:step-2 | 2. Compare a primary site

The primary carbon has only one carbon neighbor, highlighted in orange for comparison. With a small, strong nucleophile in a suitable polar aprotic solvent, bromoethane is a better SN2 candidate. An unstabilized primary carbocation is unfavorable; strong bases or heat can make elimination compete.

Displayed molecule ID: bromoethane

{"highlights": {"prin": [1], "sub": [0]}, "revealName": true}

Screenshot: substitution-step-2



redox | Organic oxidation and reduction

Track bonds to H and electronegative atoms

redox:rule | Core Rules

Oxidation of carbon usually increases bonds to O/N/halogens or decreases C–H bonds. Reduction does the reverse. C–C bonds contribute zero to carbon oxidation state. PCC under anhydrous conditions can oxidize a primary alcohol to an aldehyde. Aqueous acidic CrO3 or dichromate (Na2Cr2O7/K2Cr2O7), or suitable KMnO4 conditions, can oxidize a primary alcohol further to a carboxylic acid; substrate and conditions determine the outcome.

Visual card: Organic Oxidation States & Chemoselective Oxidizing / Reducing Reagents

Chemistry diagram: Schematic alcohol oxidation to an aldehyde under suitable conditions; not a balanced equation.

Oxidation = ↑ C–O/N/X or ↓ C–H

Reduction = ↑ C–H or ↓ C–O

Alkane < Alcohol < C=O < COOH < CO_2

1° Alcohol (RCH_2OH) | PCC → Aldehyde; $CrO_3/KMnO_4$ → COOH | $NaBH_4/LiAlH_4$ reduces back to 1° ROH

2° Alcohol (R_2CHOH) | PCC or CrO_3 → Ketone ($R_2C=O$) | No α-OH C–H left to oxidize further

3° Alcohol (R_3COH) | NO Oxidation (No carbinol C–H!) | Resists CrO_3/PCC unless C–C bonds break

Carbon Oxidation Ladder — — Alkane (−4 in CH_4) → Alcohol/Alkene → Aldehyde/Ketone → Carboxylic Acid/Derivative → CO_2 (+4).

Anhydrous PCC vs. Aqueous Jones (CrO_3/H_2SO_4) — — PCC in CH_2Cl_2 stops cleanly at the ALDEHYDE on 1° alcohols; aqueous CrO_3 or $KMnO_4$ hydrates the aldehyde and oxidizes it to –COOH.

$NaBH_4$ vs. $LiAlH_4$ Hydride Reactivity — — Mild $NaBH_4$ reduces ONLY aldehydes and ketones (plus acid chlorides); fierce $LiAlH_4$ reduces aldehydes, ketones, esters, acids, and amides!

redox:cue | Memory Cues

More bonds to O, less H: carbon is oxidized. Hydride addition to C=O generally reduces it.

Visual card: Which Oxidant or Reductant Matches Your Target Transformation?

Select your starting material and desired redox product:

1° Alcohol → Stop at Aldehyde (RCHO) — Use Anhydrous PCC (or DMP / Swern) — Without water, no geminal diol (hydrate) can form, so oxidation halts cleanly at the aldehyde stage.

1° Alcohol or Aldehyde → Carboxylic Acid (RCOOH) — Use Jones (CrO_3/H_2SO_4, H_2O) or $KMnO_4$ — Water forms the geminal diol $RCH(OH)_2$, which has a second C–H bond for chromate to oxidize to –COOH.

Reduce Ketone/Aldehyde in Presence of an Ester — Use Mild $NaBH_4$ in MeOH/EtOH — $NaBH_4$ selectively reduces the more electrophilic ketone/aldehyde while leaving the ester untouched!

redox:trap | Common Mistakes / MCAT Traps

NaBH4 commonly reduces aldehydes and ketones but ordinarily not esters or carboxylic acids. LiAlH4 reduces esters and acids to alcohols and amides to amines, using suitable nonaqueous reaction conditions followed by a separate workup. Reagent, solvent and workup matter; do not treat LiAlH4 as compatible with an aqueous reaction medium.

Visual card: Oxidizing a 3° Alcohol with CrO_3 or Reducing an Ester with $NaBH_4$

Common Trap — — Drawing a ketone product from a 3° alcohol + CrO_3, or using $NaBH_4$ to reduce a carboxylic acid or ester.

Correct Reasoning — — Oxidation of an alcohol to C=O requires breaking a C–H bond on the carbinol carbon (3° alcohols have zero!), and esters/acids require $LiAlH_4$.

redox:app | MCAT Applications

NAD+/NADH couples biological alcohol/carbonyl oxidation states through hydride transfer; not every laboratory reagent has a direct biological analogue.

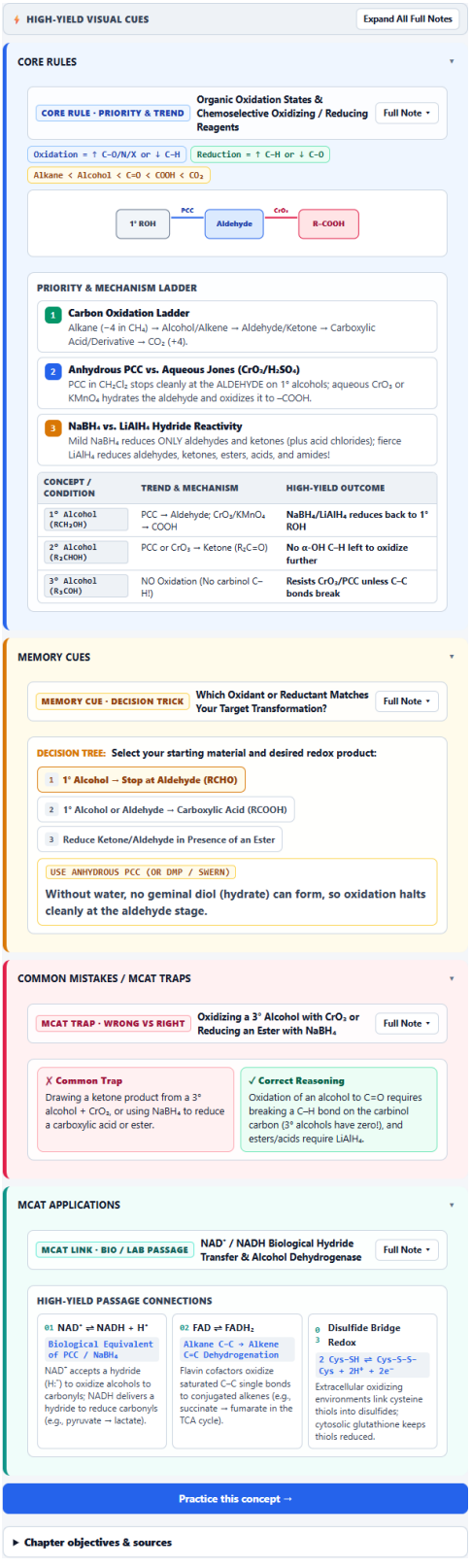
Visual card: NAD^+ / NADH Biological Hydride Transfer & Alcohol Dehydrogenase

$NAD^+ \rightleftharpoons NADH + H^+$ — Biological Equivalent of PCC / $NaBH_4$ — NAD^+ accepts a hydride ($H:^-$) to oxidize alcohols to carbonyls; NADH delivers a hydride to reduce carbonyls (e.g., pyruvate → lactate).

$FAD \rightleftharpoons FADH_2$ — Alkane C–C → Alkene C=C Dehydrogenation — Flavin cofactors oxidize saturated C–C single bonds to conjugated alkenes (e.g., succinate → fumarate in the TCA cycle).

Disulfide Bridge Redox — 2 Cys–SH ⇌ Cys–S–S–Cys + $2H^+ + 2e^-$ — Extracellular oxidizing environments link cysteine thiols into disulfides; cytosolic glutathione keeps thiols reduced.

Screenshot: redox-visuals



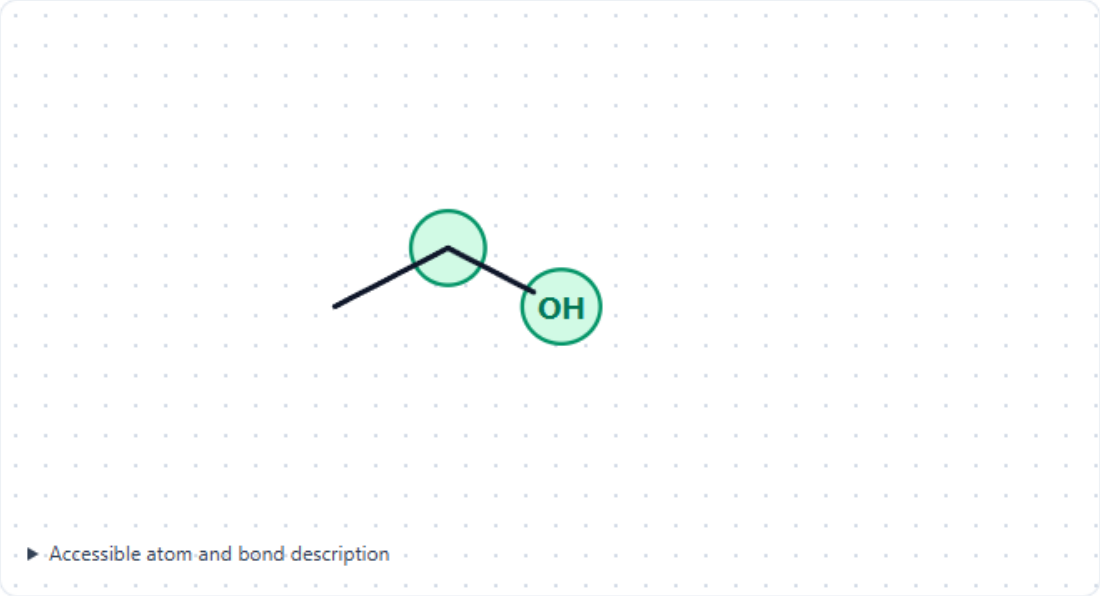
redox:step-1 | 1. Alcohol carbon

For the CH2OH carbon, one C–O bond contributes +1 and two C–H bonds contribute −2: oxidation state −1.

Displayed molecule ID: ethanol

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: redox-step-1



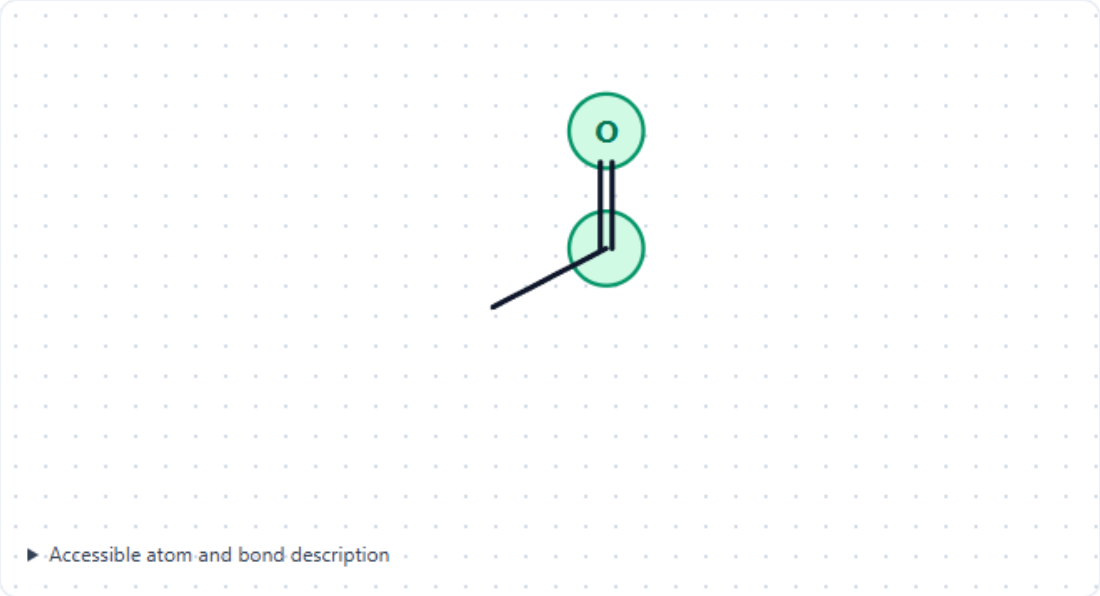
redox:step-2 | 2. Aldehyde carbon

For CHO, C=O contributes +2, C–H contributes −1, and C–C contributes 0: oxidation state +1. Alcohol → aldehyde is a two-electron oxidation.

Displayed molecule ID: acetaldehyde

{"highlights": {"prin": [1, 2]}, "revealName": true}

Screenshot: redox-step-2



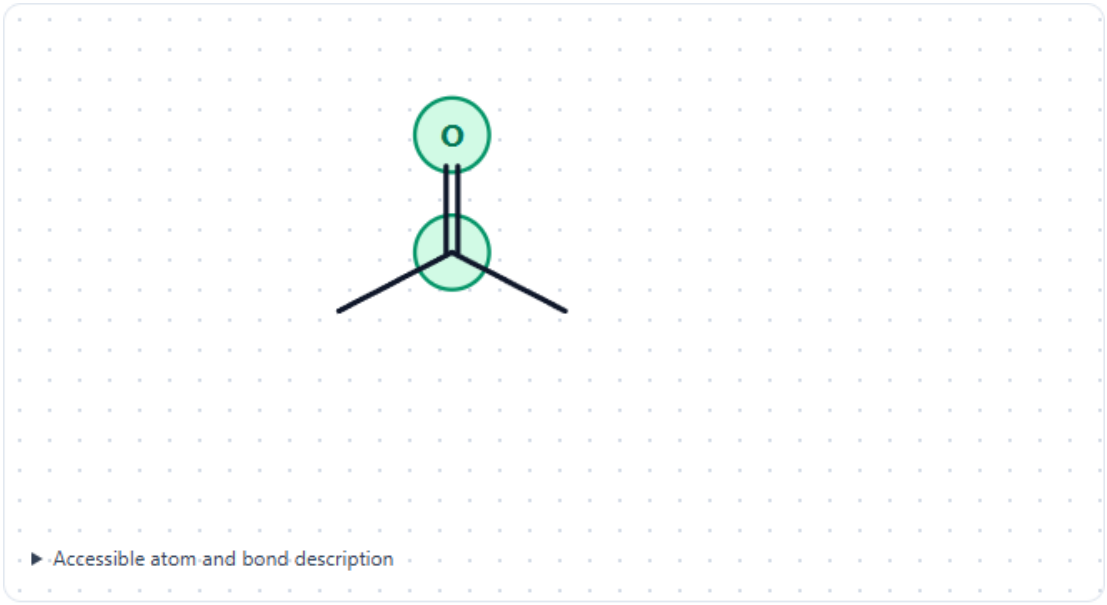
redox:step-3 | 3. Ketone reduction

Hydride adds to the electrophilic carbonyl carbon; protonation of oxygen gives a secondary alcohol.

Displayed molecule ID: acetone

{"highlights": {"prin": [1, 3]}, "revealName": true}

Screenshot: redox-step-3



selectivity | Chemoselectivity and a reaction workflow

Choose by reactivity, not naming seniority

selectivity:rule | Core Rules

Inventory functional groups, identify reagent roles, locate the likely reactive site, account for solvent and acid–base steps, then track bonds and stereochemistry. Chemoselectivity means preferential reaction at one functional group.

Visual card: Chemo-, Regio-, and Stereoselectivity & The SN1/SN2/E1/E2 Competition Grid

Chemistry diagram: Two functional-group foci on the same molecular graph, not a reactant/product pair.

Chemoselective = Which FG

Regioselective = Where on FG

Stereoselective = Which 3D Isomer

Chemoselectivity | Selects 1 functional group over another | $NaBH_4$ reduces ketone but spares ester

Regioselectivity | Selects 1 constitutional site | Zaitsev (internal) vs Hofmann (terminal) alkene

Stereoselectivity | Favors 1 stereoisomer | E2 anti-periplanar → E-alkene > Z-alkene

Chemoselectivity (Functional Group Choice) — — Exploits intrinsic reactivity differences (e.g., Aldehyde > Ketone > Ester toward nucleophiles) or uses protecting groups.

Regioselectivity (Constitutional Site Choice) — — Determines which positional isomer forms (e.g., small base EtO^- gives Zaitsev internal alkene; bulky t-BuO^- gives Hofmann less-substituted alkene).

Stereoselectivity & Stereospecificity — — Stereoselective reactions favor the more stable stereoisomer (E > Z); stereospecific reactions map starting geometry directly to product (SN2 inversion, anti-E2).

selectivity:cue | Memory Cues

Group inventory → reagent → first step → bond changes → product checks.

Visual card: Master Substitution vs. Elimination Decision Engine

Given an alkyl halide RX, how do you predict SN2 vs SN1 vs E2 vs E1?

Strong Unhindered Nucleophile (CN^-, RS^-, I^-) + 1°/2° RX — SN2 Dominates (Backside Inversion) — Low basicity + minimal steric hindrance attacks the α-carbon σ* orbital cleanly.

Strong Base (HO^-, RO^- on 2°/3° RX) OR Bulky Base (t-BuO^-, LDA) — E2 Dominates (Concerted Anti-Periplanar) — Strong bases abstract a β-proton; small bases give the more substituted Zaitsev alkene, while t-BuO^- gives the Hofmann alkene.

Weak Base/Nu (H_2O, ROH) + 3° RX — SN1 + E1 Mixture (Heat Favors E1) — Both share the same rate-determining carbocation step; higher temperature (Δ) favors E1 via entropy (TΔS).

selectivity:trap | Common Mistakes / MCAT Traps

Do not substitute a naming-priority or carbon-oxidation-state ranking for reaction analysis. An ester is senior to a ketone in naming, yet its alkoxy oxygen donates electron density into the carbonyl by resonance. Under ordinary NaBH4 conditions, the ketone can be reduced while the ester survives. An amide is also resonance-stabilized. Acid–base reactions, activation and conditions can change which site reacts first.

Visual card: Confusing Regioselectivity with Chemoselectivity or Stereoselectivity

Common Trap — — Calling Markovnikov vs anti-Markovnikov addition "chemoselective" or "stereoselective".

Correct Reasoning — — Different atom connectivity (1-ol vs 2-ol, or internal vs terminal alkene) is REGIOSELECTIVE; different 3D arrangement (R/S or E/Z) is STEREOSELECTIVE.

selectivity:app | MCAT Applications

Acetals protect aldehyde/ketone carbonyls from many basic and nucleophilic conditions; aqueous acid restores the carbonyl. Protection replaces the carbonyl pi system with a different functional group, changing electronic reactivity rather than merely adding steric bulk. Carbonyl compounds may also react through acidic alpha hydrogens: deprotonation creates an enolate nucleophile instead of nucleophilic addition at the carbonyl carbon.

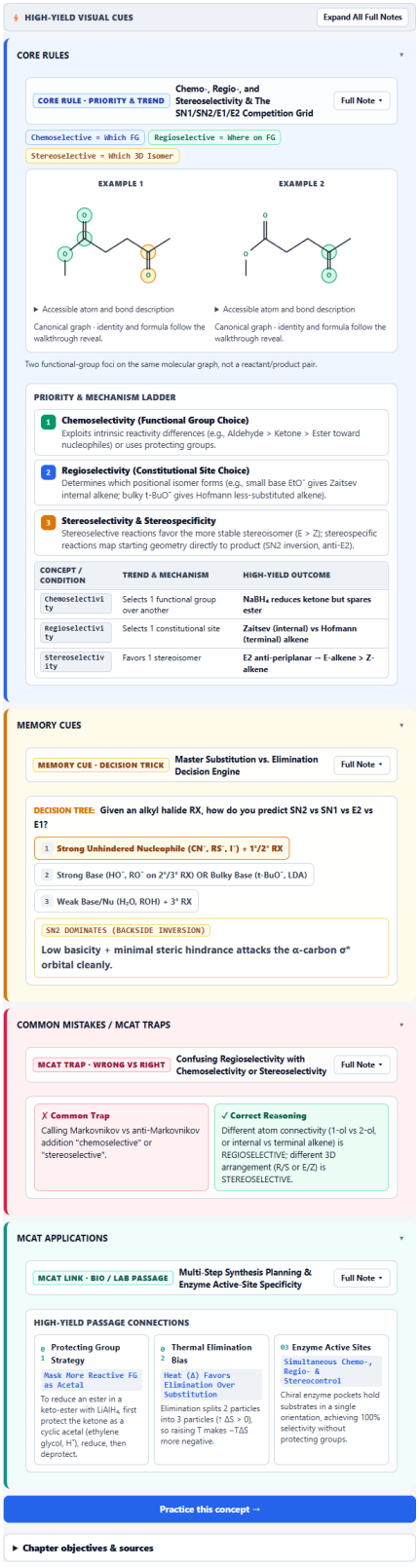
Visual card: Multi-Step Synthesis Planning & Enzyme Active-Site Specificity

Protecting Group Strategy — Mask More Reactive FG as Acetal — To reduce an ester in a keto-ester with $LiAlH_4$, first protect the ketone as a cyclic acetal (ethylene glycol, H^+), reduce, then deprotect.

Thermal Elimination Bias — Heat (Δ) Favors Elimination Over Substitution — Elimination splits 2 particles into 3 particles (↑ $\Delta S > 0$), so raising T makes $-T\Delta S$ more negative.

Enzyme Active Sites — Simultaneous Chemo-, Regio- & Stereocontrol — Chiral enzyme pockets hold substrates in a single orientation, achieving 100% selectivity without protecting groups.

Screenshot: selectivity-visuals



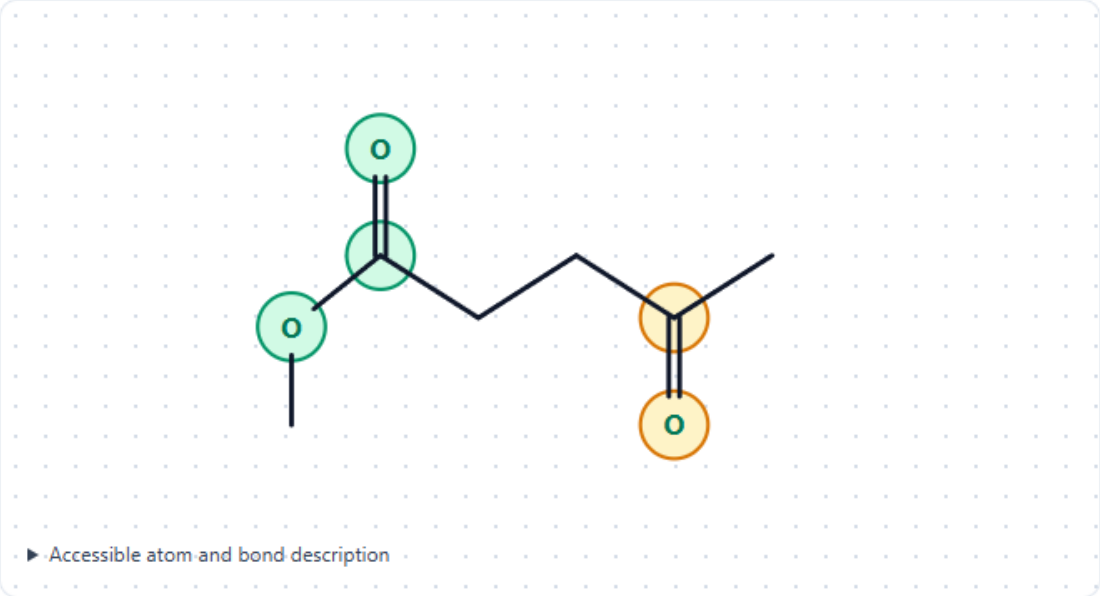
selectivity:step-1 | 1. Inventory two carbonyls

This molecule contains an ester at the left and a ketone at the right. The ester has an oxygen substituent directly on its carbonyl carbon.

Displayed molecule ID: ketoester

{"highlights": {"prin": [0, 5, 6], "sub": [3, 8]}, "revealName": true}

Screenshot: selectivity-step-1



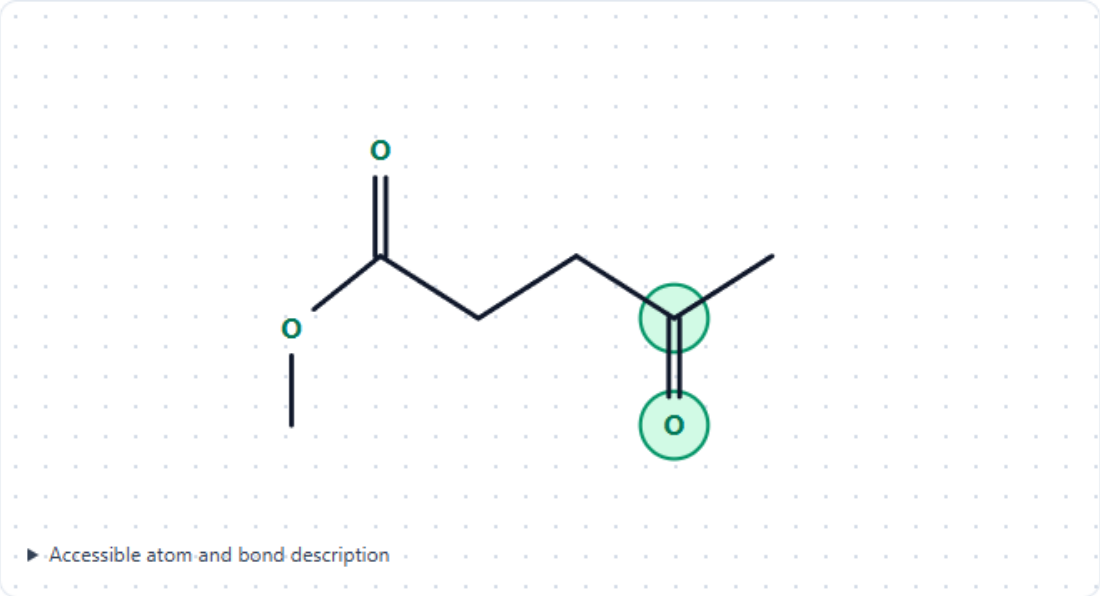
selectivity:step-2 | 2. Predict a selective step

Ordinary NaBH4 conditions favor reduction of the ketone. Ester resonance donation lowers its electrophilicity relative to the ketone; a stronger reducing system can change the outcome.

Displayed molecule ID: ketoester

{"highlights": {"prin": [3, 8]}, "revealName": false}

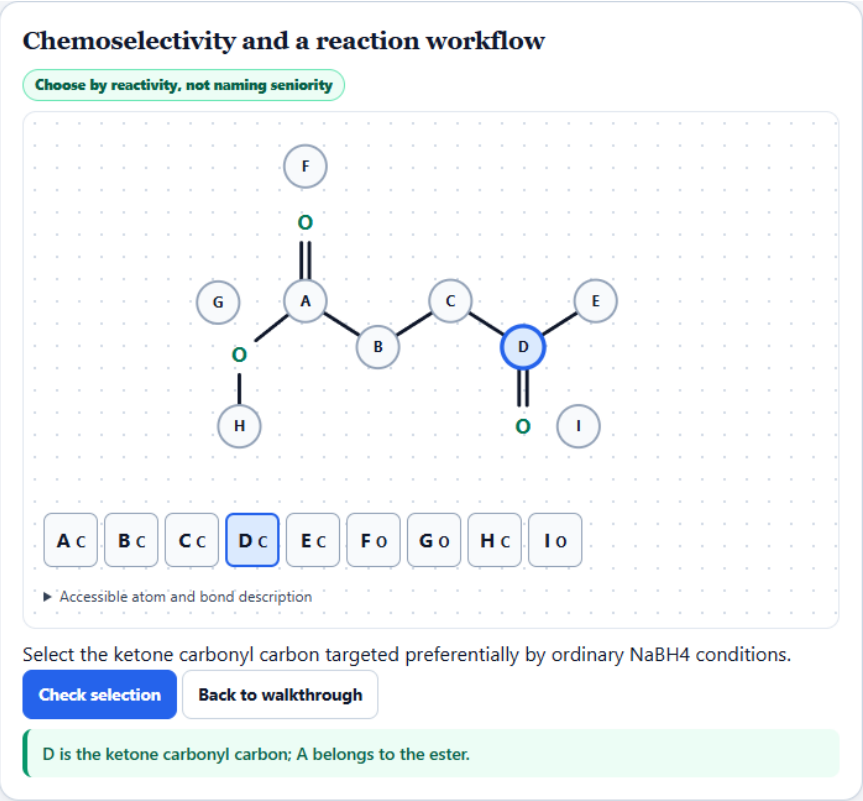
Screenshot: selectivity-step-2



Embedded challenge: Select the ketone carbonyl carbon targeted preferentially by ordinary NaBH4 conditions.

{"prompt": "Select the ketone carbonyl carbon targeted preferentially by ordinary NaBH4 conditions.", "answer": [3], "explanation": "D is the ketone carbonyl carbon; A belongs to the ester."}

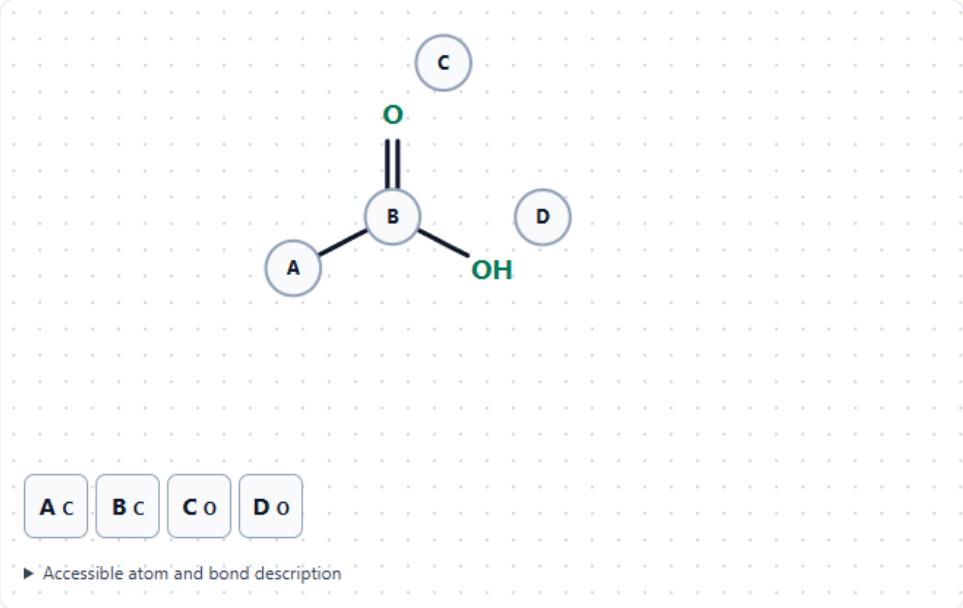
Screenshot: selectivity-step-2-challenge



pka-equilibrium | Estimate acid–base equilibrium

The displayed carboxyl group is a structural example for Stage 1. Stages 2–3 use a separate hypothetical proton-transfer equilibrium, HA + B ⇌ A− + BH+, with all pKa values on a comparable solvent/temperature scale; they are not measured values for the displayed ethanoic acid.

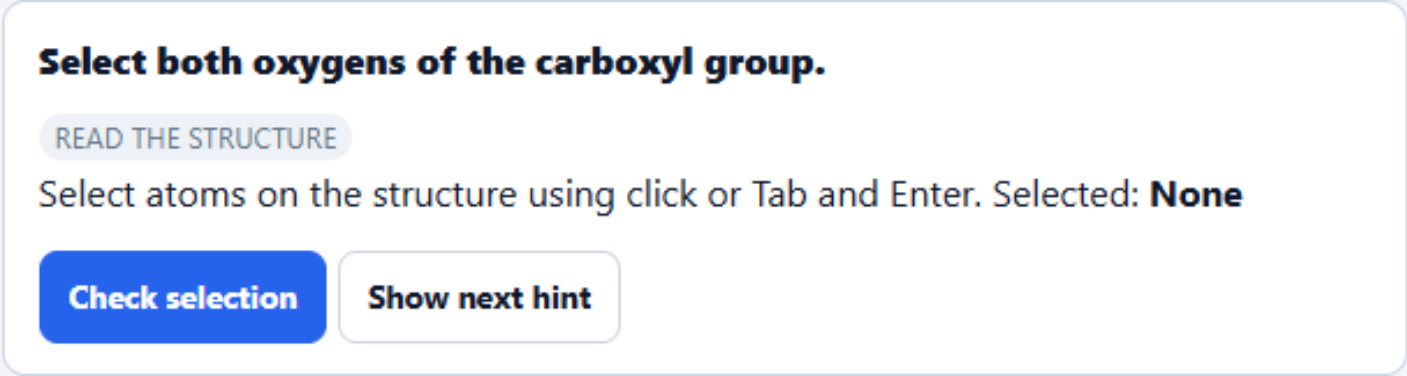
Screenshot: pka-equilibrium-stage-1-molecule



pka-equilibrium:stage-1 | Select both oxygens of the carboxyl group.

Answer type: atoms

Screenshot: pka-equilibrium-stage-1-question



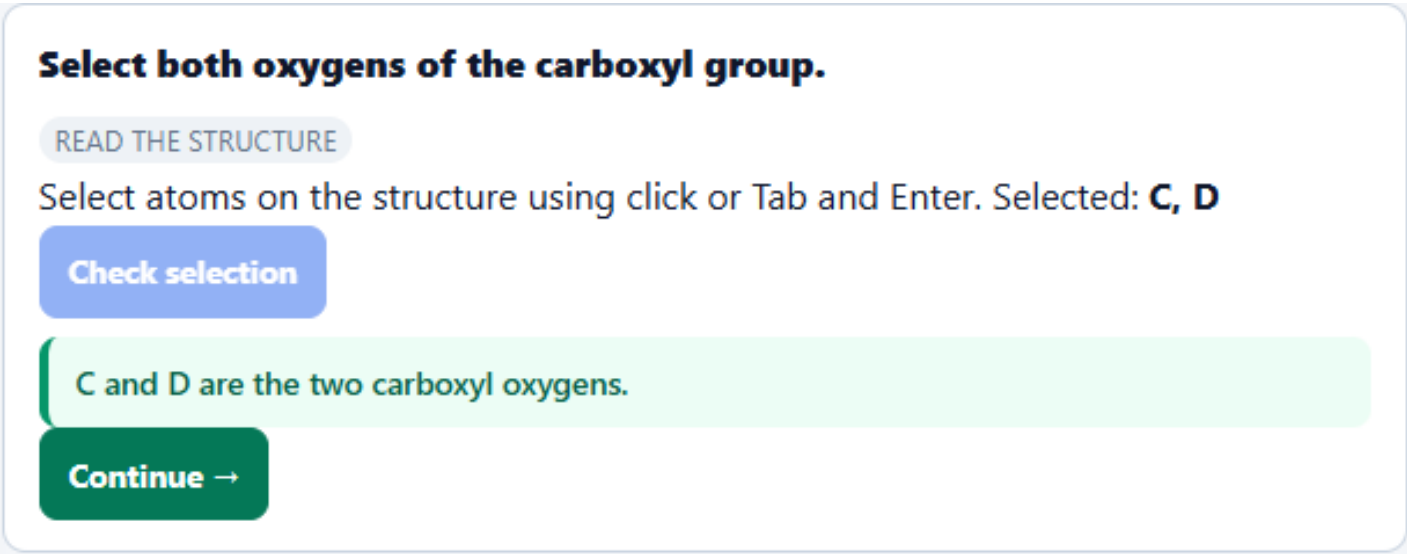
{"authorAnswer": [2, 3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [2, 3]}}

Hint 1: Locate C=O and O–H on the same carbon.

Hint 2: Select the two O labels.

Explanation: C and D are the two carboxyl oxygens.

Screenshot: pka-equilibrium-stage-1-feedback



pka-equilibrium:stage-2 | For HA + B ⇌ A− + BH+, if pKa(HA)=5 and pKa(BH+)=10, equilibrium favors...

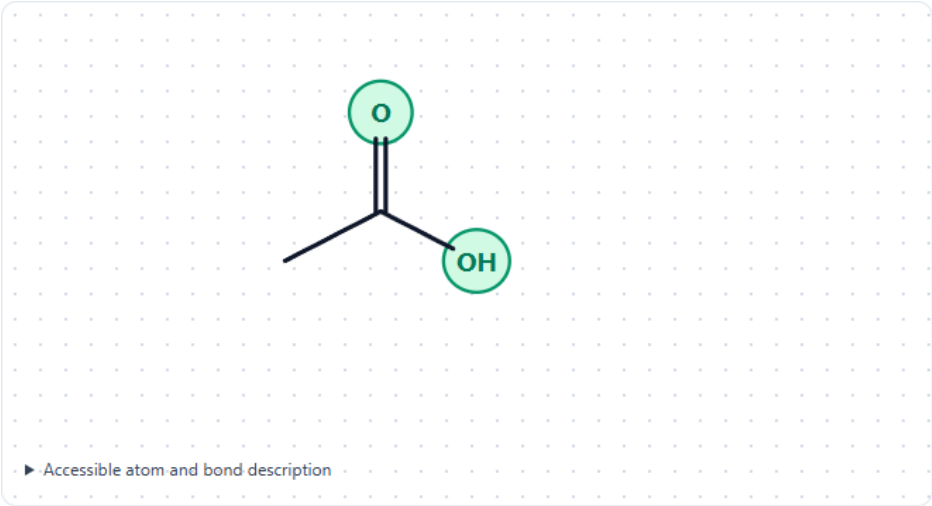
Answer type: choice

Option 0 (A): Reactants

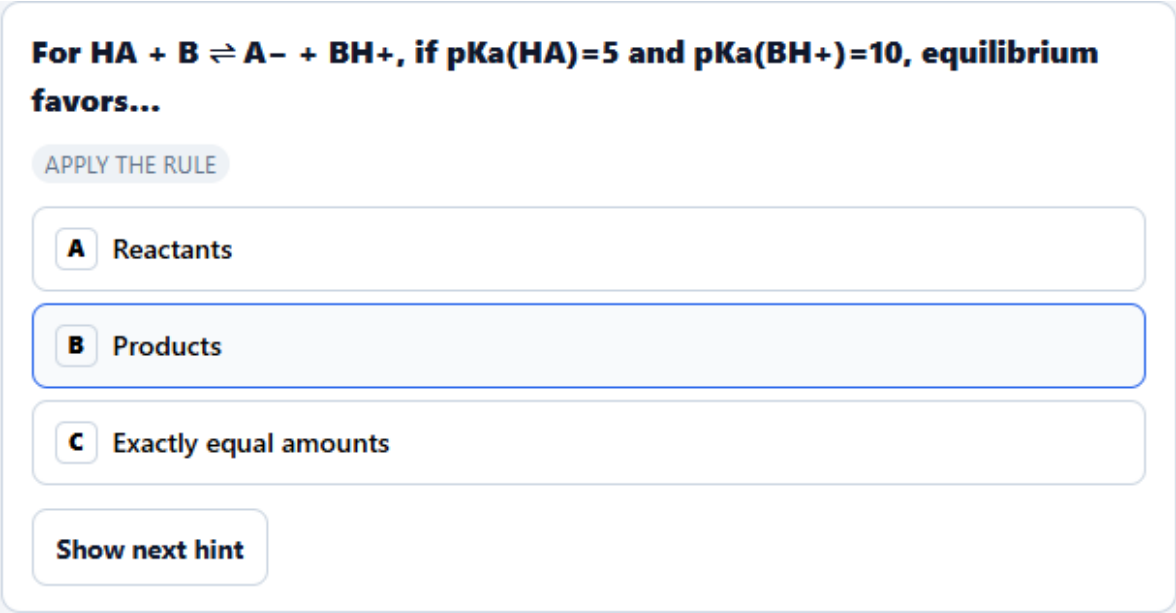
Option 1 (B): Products

Option 2 (C): Exactly equal amounts

Screenshot: pka-equilibrium-stage-2-molecule



Screenshot: pka-equilibrium-stage-2-question



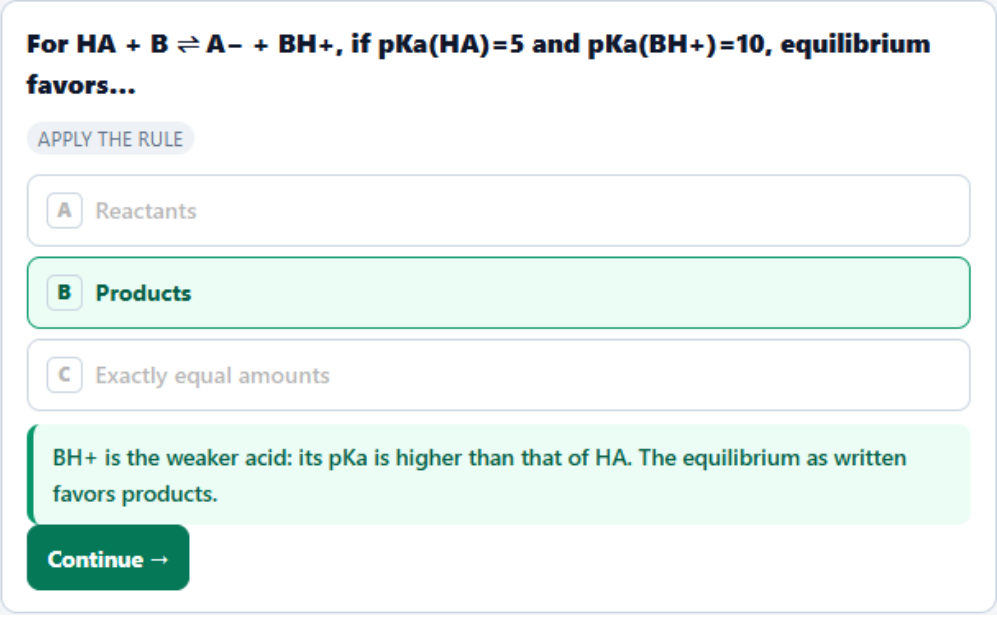
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Compare the two acids.

Hint 2: Equilibrium favors the side with the higher-pKa acid.

Explanation: BH+ is the weaker acid: its pKa is higher than that of HA. The equilibrium as written favors products.

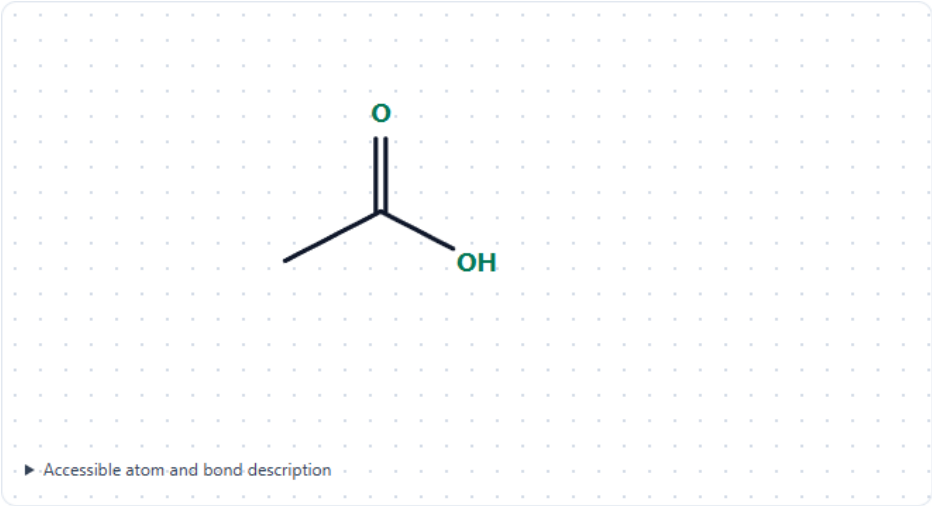
Screenshot: pka-equilibrium-stage-2-feedback



pka-equilibrium:stage-3 | For that forward equilibrium, using pKa(HA)=5 and pKa(BH+)=10, estimate K.

Answer type: number | unit: dimensionless

Screenshot: pka-equilibrium-stage-3-molecule



Screenshot: pka-equilibrium-stage-3-question

For that forward equilibrium, using pKa(HA)=5 and pKa(BH+)=10, estimate K.

CALCULATE AND INTERPRET

Numeric answer (dimensionless)

Check answer

Show next hint

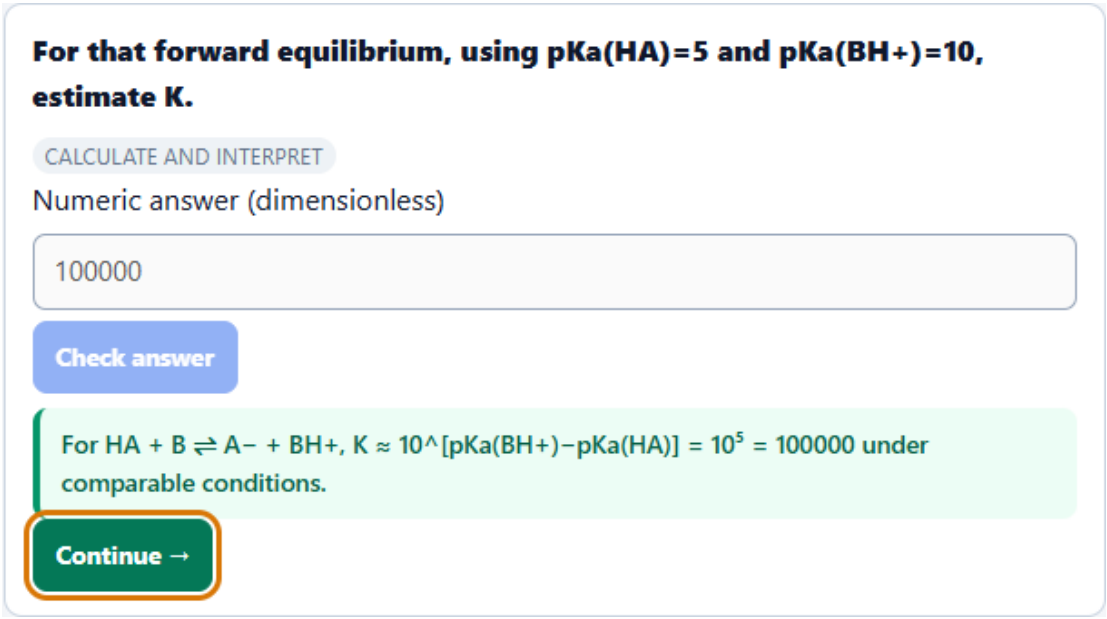
{"authorAnswer": 100000, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Subtract reactant-acid pKa from product-acid pKa.

Hint 2: Raise ten to that difference.

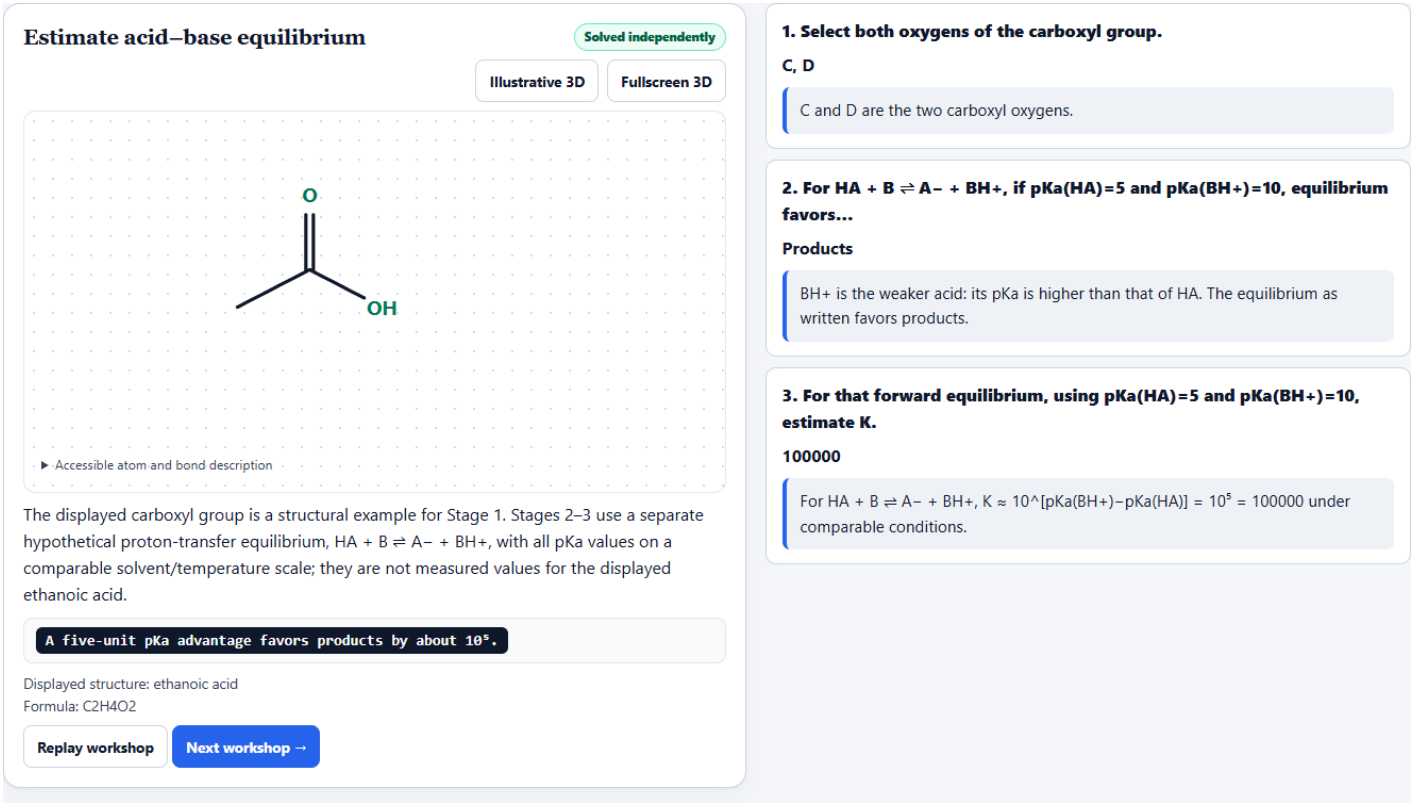
Explanation: For HA + B ⇌ A− + BH+, K ≈ 10^[pKa(BH+)−pKa(HA)] = 10^5 = 100000 under comparable conditions.

Screenshot: pka-equilibrium-stage-3-feedback



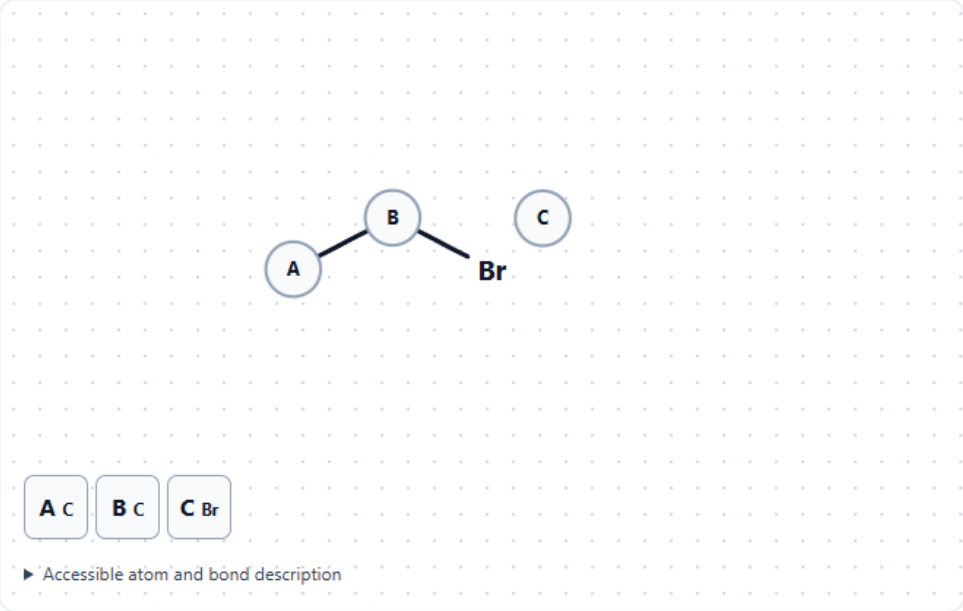
Completion summary: A five-unit pKa advantage favors products by about 10^5.

Screenshot: pka-equilibrium-complete



identify-partners | Find the substitution partners

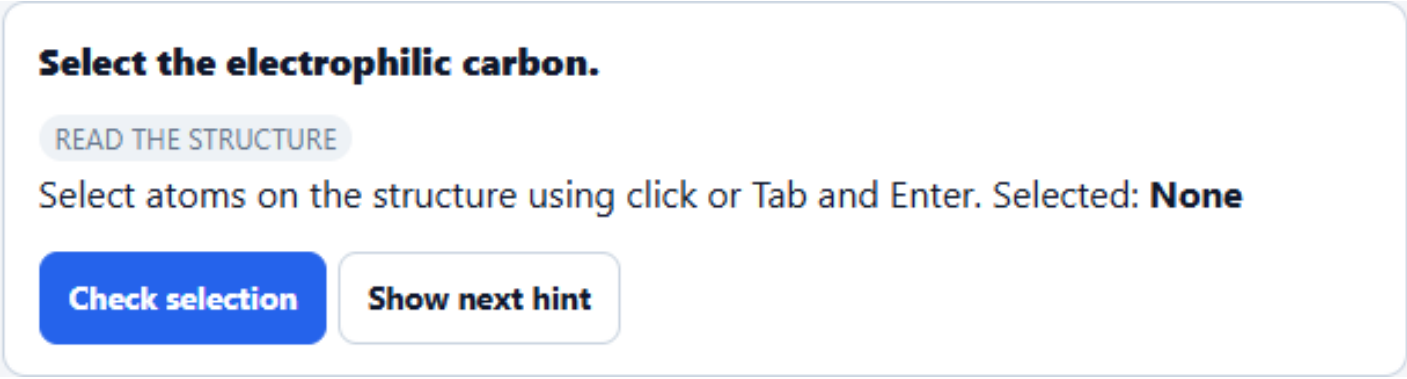
Screenshot: identify-partners-stage-1-molecule



identify-partners:stage-1 | Select the electrophilic carbon.

Answer type: atoms

Screenshot: identify-partners-stage-1-question



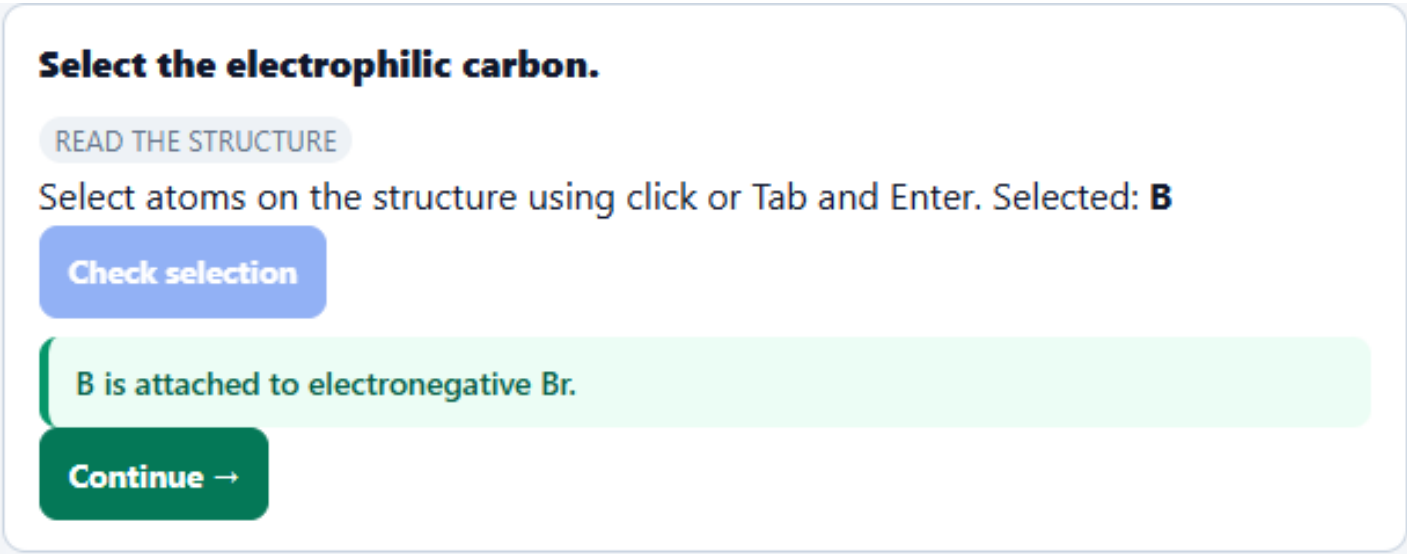
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1]}}

Hint 1: Look for the carbon bonded to the leaving group.

Hint 2: The polarized C–Br bond identifies it.

Explanation: B is attached to electronegative Br.

Screenshot: identify-partners-stage-1-feedback



identify-partners:stage-2 | Among these choices (including a reagent not drawn), which species supplies a lone pair as a nucleophile?

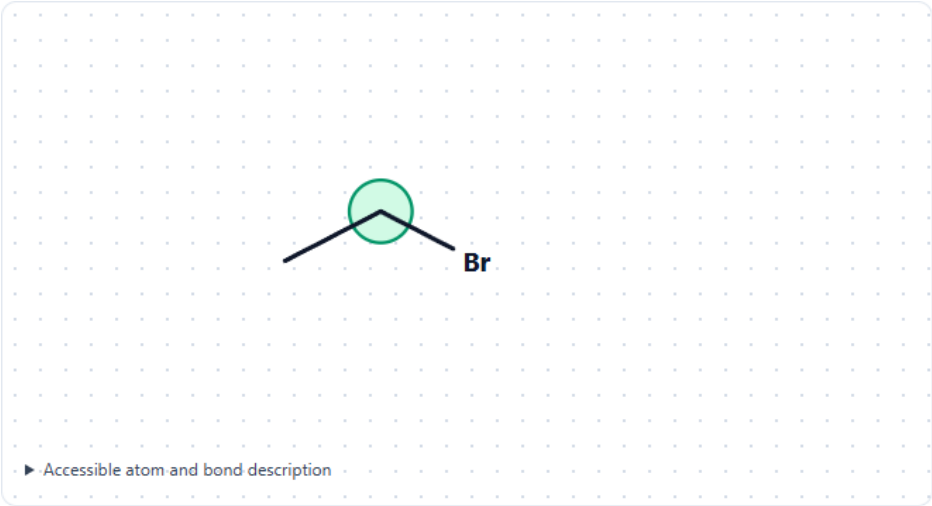
Answer type: choice

Option 0 (A): Hydroxide ion

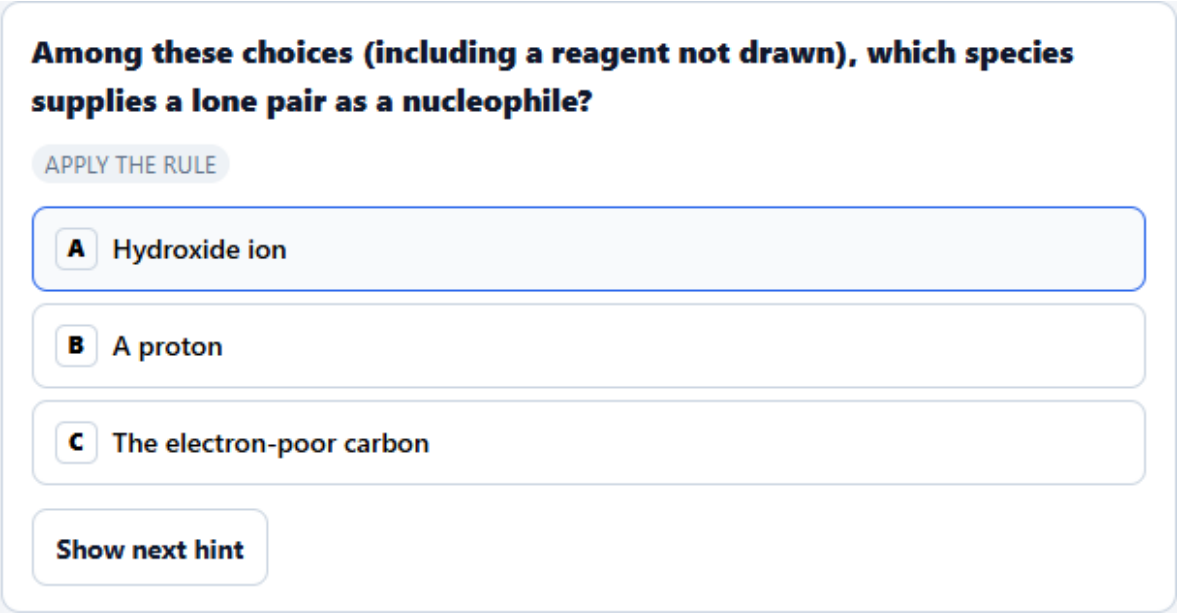
Option 1 (B): A proton

Option 2 (C): The electron-poor carbon

Screenshot: identify-partners-stage-2-molecule



Screenshot: identify-partners-stage-2-question



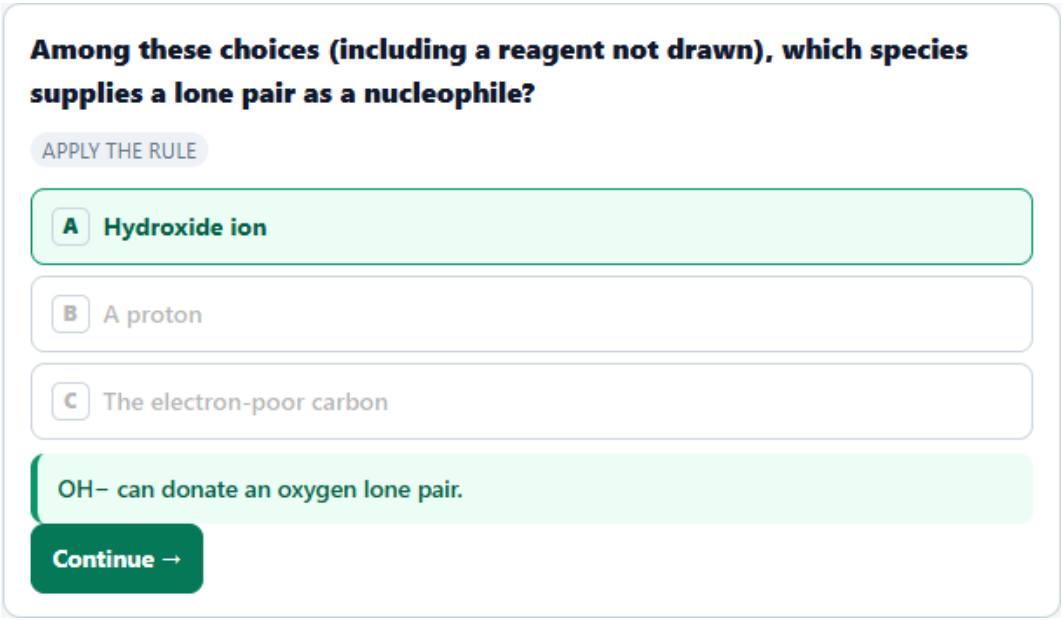
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Nucleophiles are electron-pair donors.

Hint 2: A negative charge often indicates available electrons.

Explanation: OH− can donate an oxygen lone pair.

Screenshot: identify-partners-stage-2-feedback



identify-partners:stage-3 | When C–Br breaks heterolytically, the pair goes to…

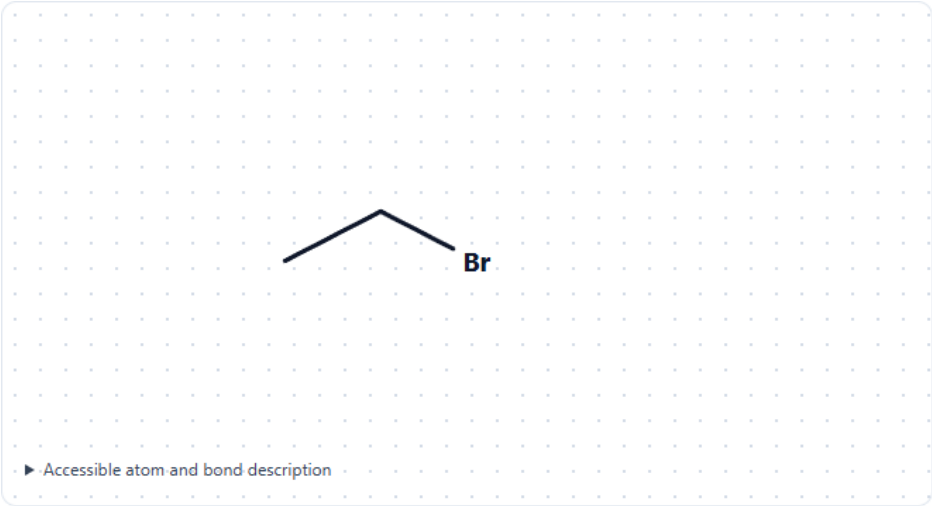
Answer type: choice

Option 0 (A): Carbon

Option 1 (B): Bromine

Option 2 (C): Neither atom

Screenshot: identify-partners-stage-3-molecule



Screenshot: identify-partners-stage-3-question



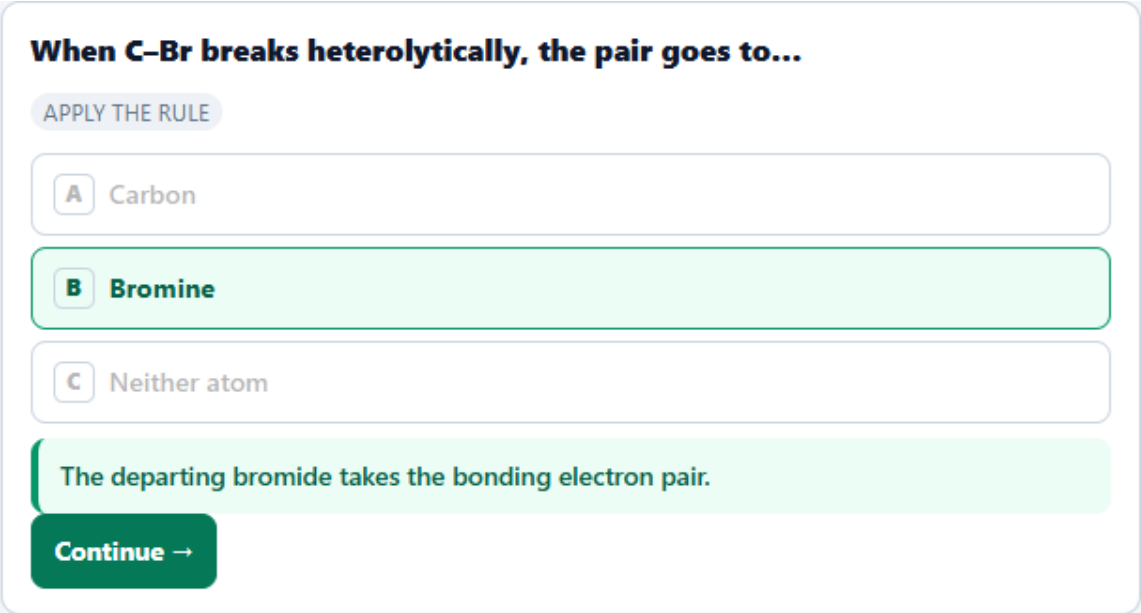
{"authorAnswer": 1, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: A leaving group retains the bond electrons.

Hint 2: Bromide is a relatively stable anion.

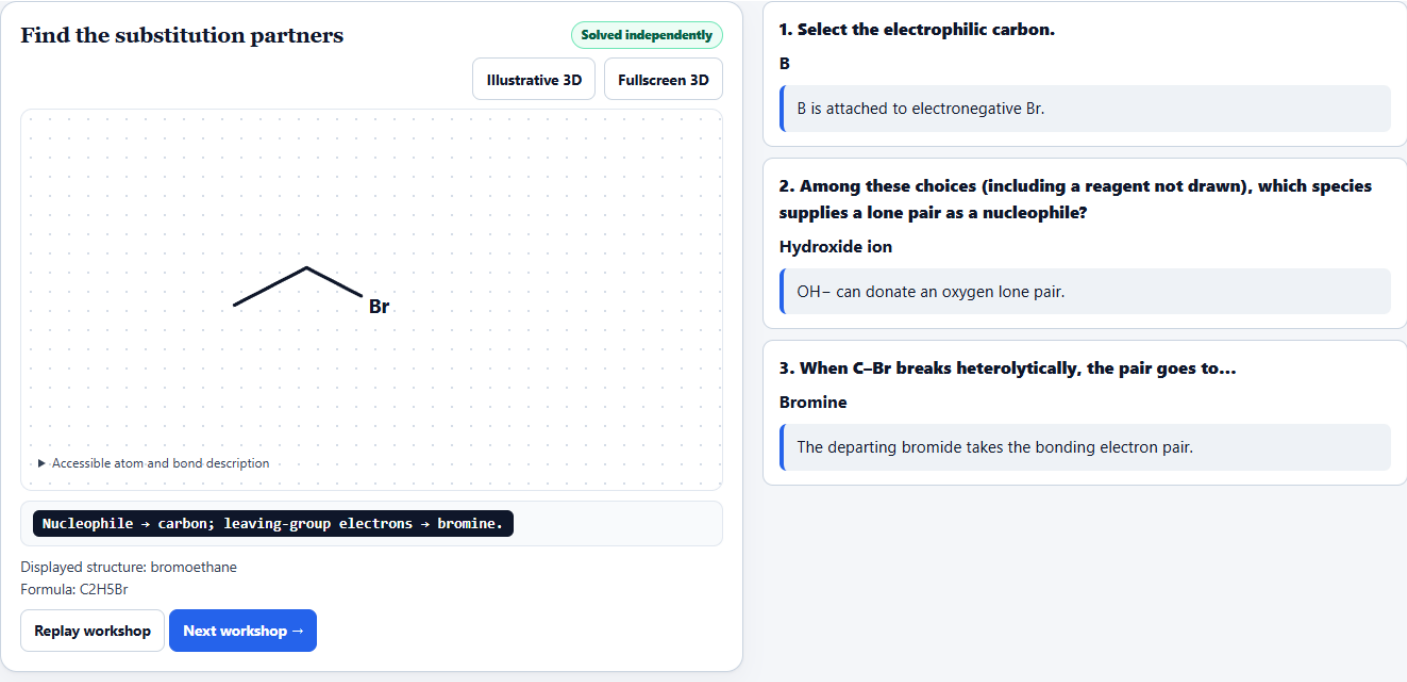
Explanation: The departing bromide takes the bonding electron pair.

Screenshot: identify-partners-stage-3-feedback



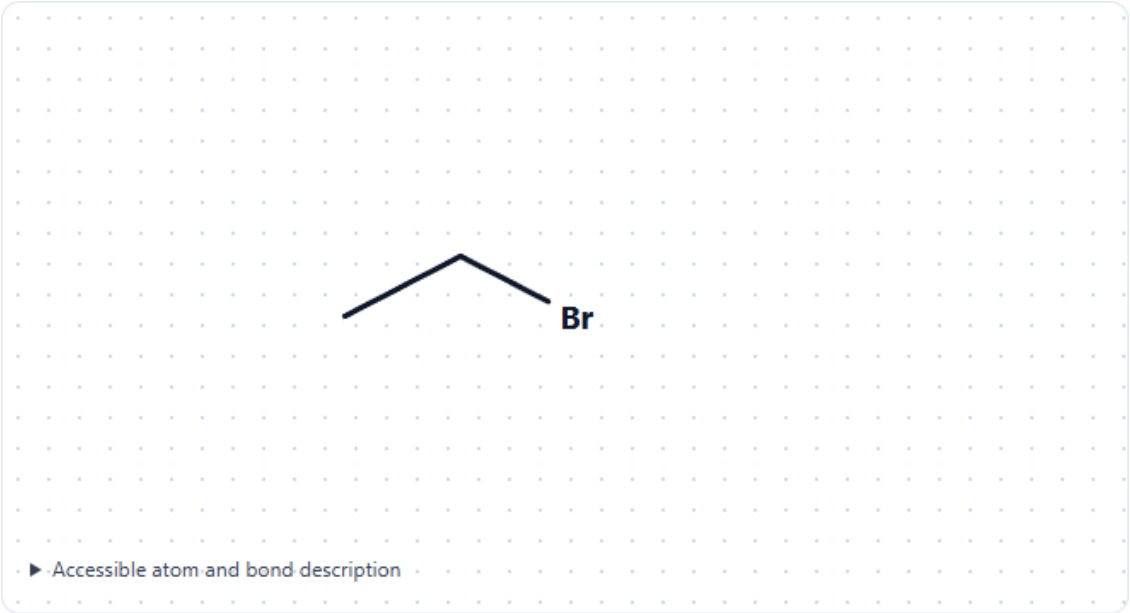
Completion summary: Nucleophile → carbon; leaving-group electrons → bromine.

Screenshot: identify-partners-complete



sn2-rate | Mechanism and kinetics

Screenshot: sn2-rate-stage-1-molecule



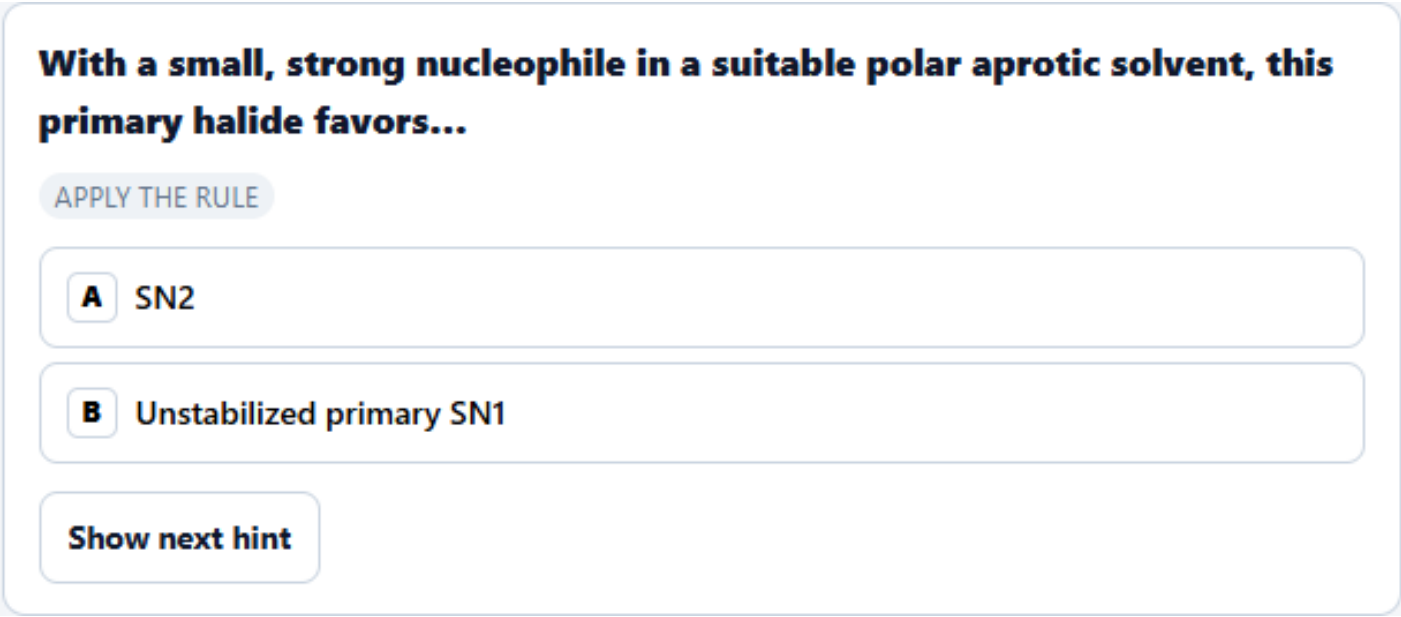
sn2-rate:stage-1 | With a small, strong nucleophile in a suitable polar aprotic solvent, this primary halide favors...

Answer type: choice

Option 0 (A): SN2

Option 1 (B): Unstabilized primary SN1

Screenshot: sn2-rate-stage-1-question



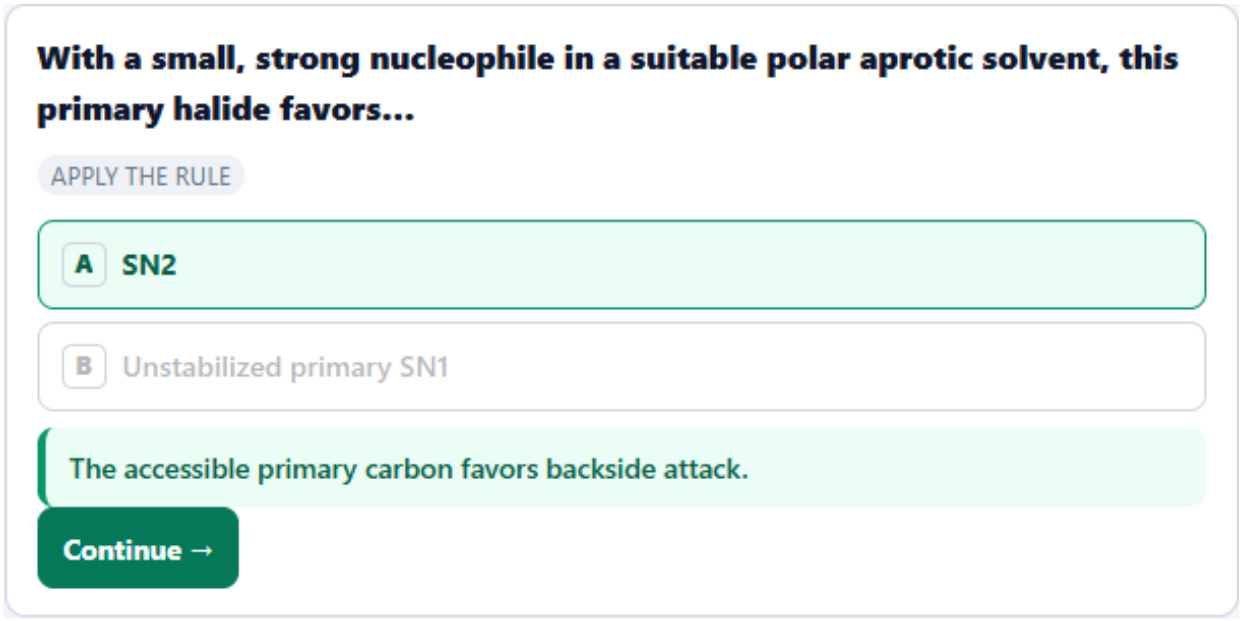
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Assess steric access and carbocation stability.

Hint 2: An unstabilized primary carbocation is poor.

Explanation: The accessible primary carbon favors backside attack.

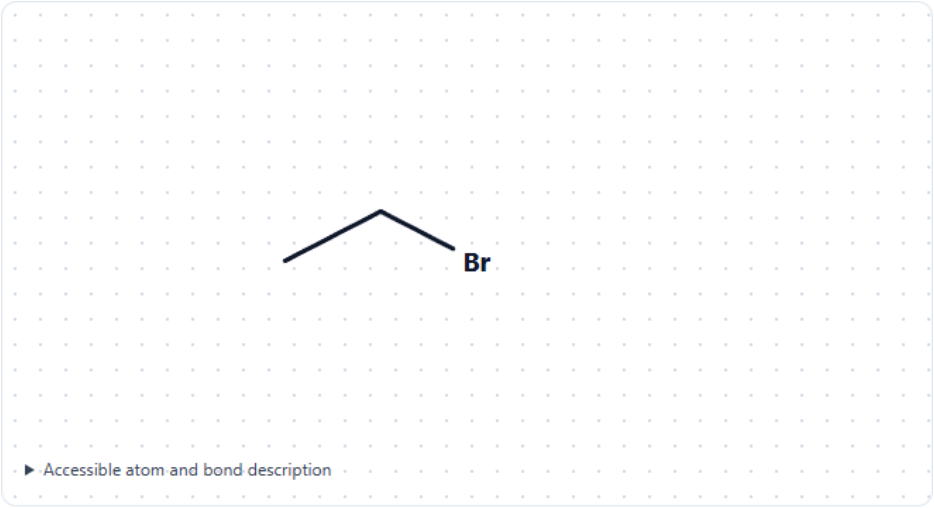
Screenshot: sn2-rate-stage-1-feedback



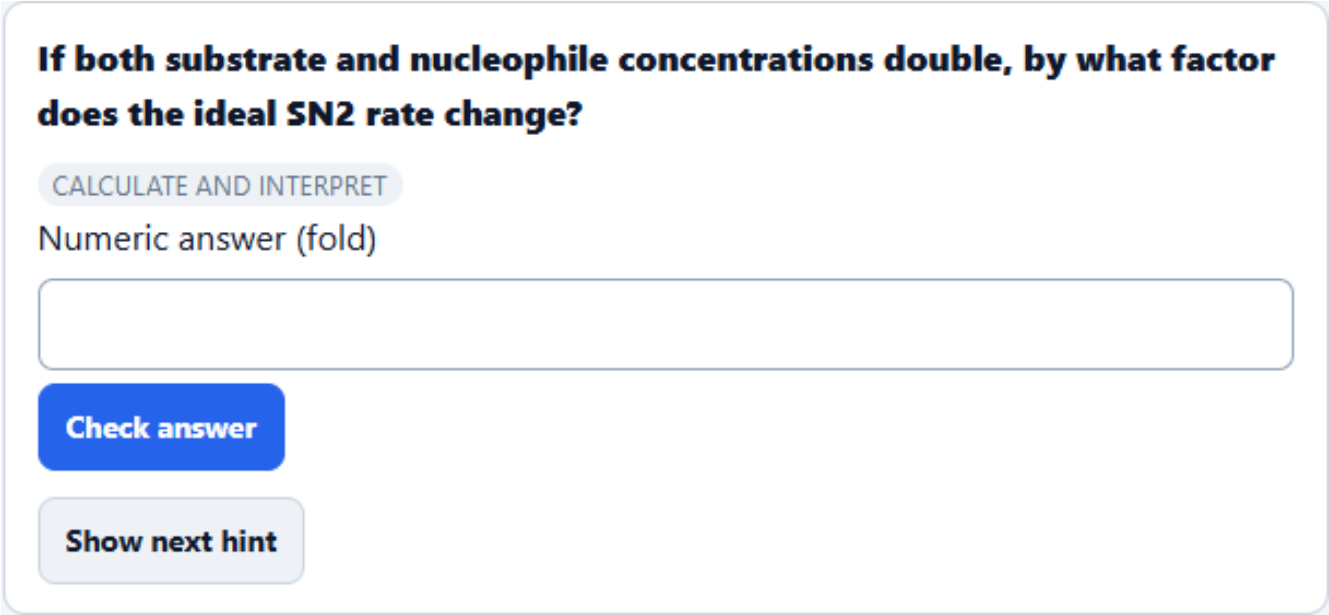
sn2-rate:stage-2 | If both substrate and nucleophile concentrations double, by what factor does the ideal SN2 rate change?

Answer type: number | unit: fold

Screenshot: sn2-rate-stage-2-molecule



Screenshot: sn2-rate-stage-2-question



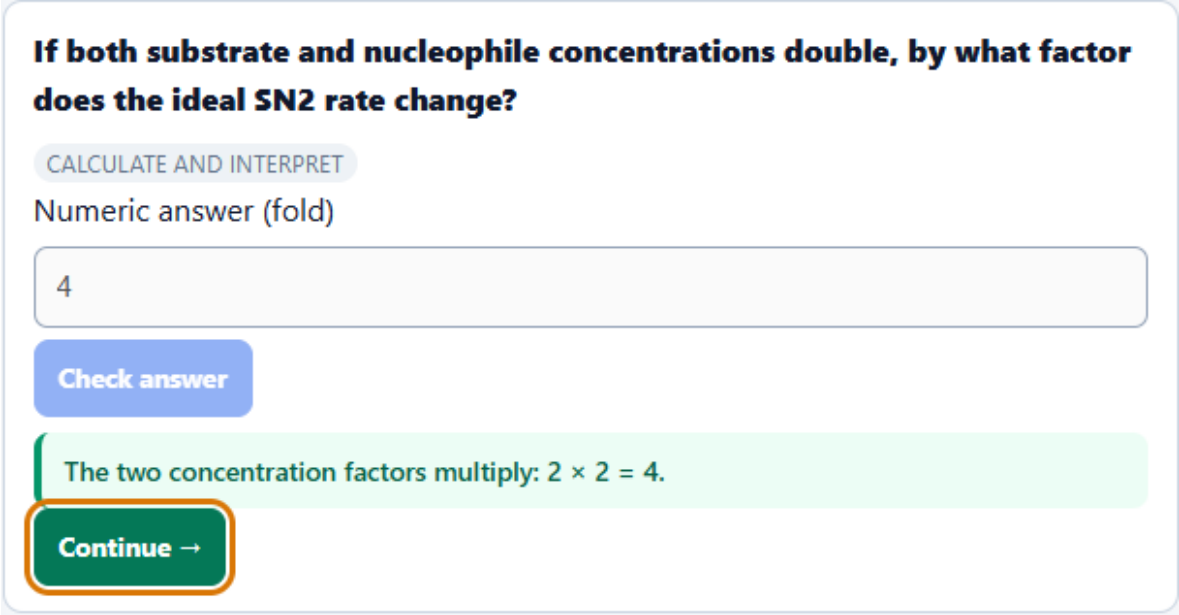
{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use rate = k[substrate][nucleophile].

Hint 2: Each concentration contributes one factor of two.

Explanation: The two concentration factors multiply: 2 × 2 = 4.

Screenshot: sn2-rate-stage-2-feedback



sn2-rate:stage-3 | Apply the general rule to a separate chiral SN2 center (the displayed bromoethane is achiral): its geometry undergoes...

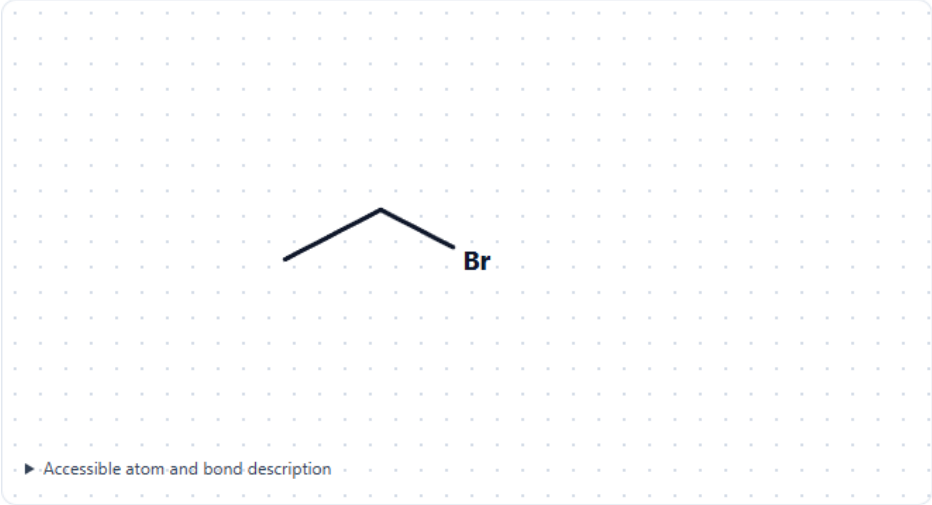
Answer type: choice

Option 0 (A): Inversion

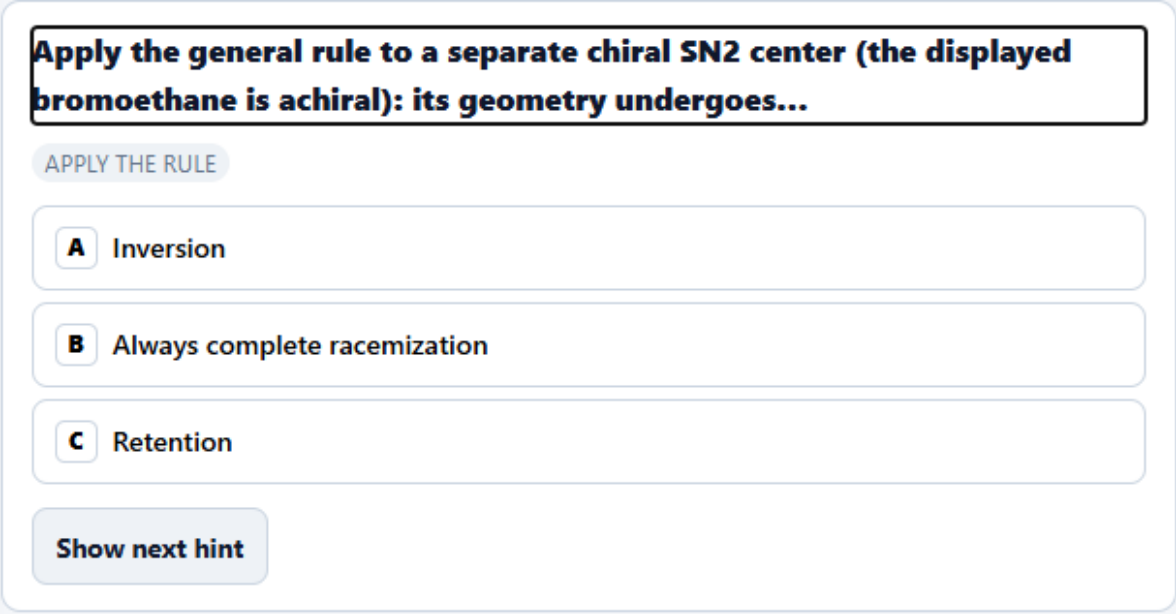
Option 1 (B): Always complete racemization

Option 2 (C): Retention

Screenshot: sn2-rate-stage-3-molecule



Screenshot: sn2-rate-stage-3-question



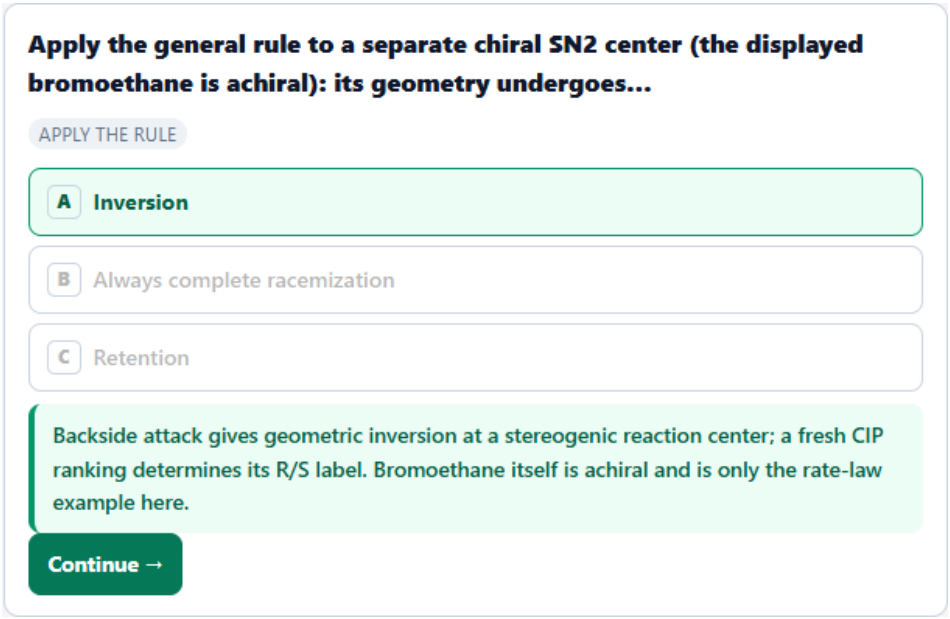
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: The nucleophile approaches opposite the leaving group.

Hint 2: Think of an umbrella turning inside out.

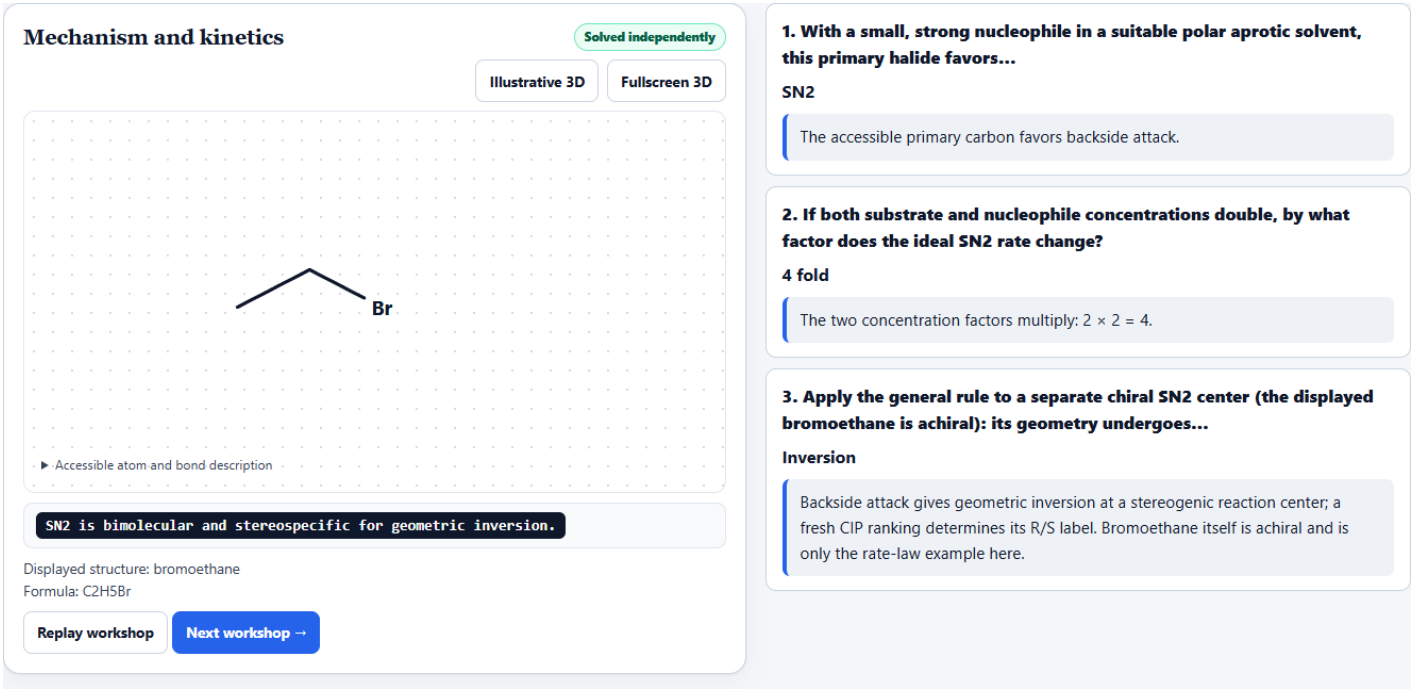
Explanation: Backside attack gives geometric inversion at a stereogenic reaction center; a fresh CIP ranking determines its R/S label. Bromoethane itself is achiral and is only the rate-law example here.

Screenshot: sn2-rate-stage-3-feedback



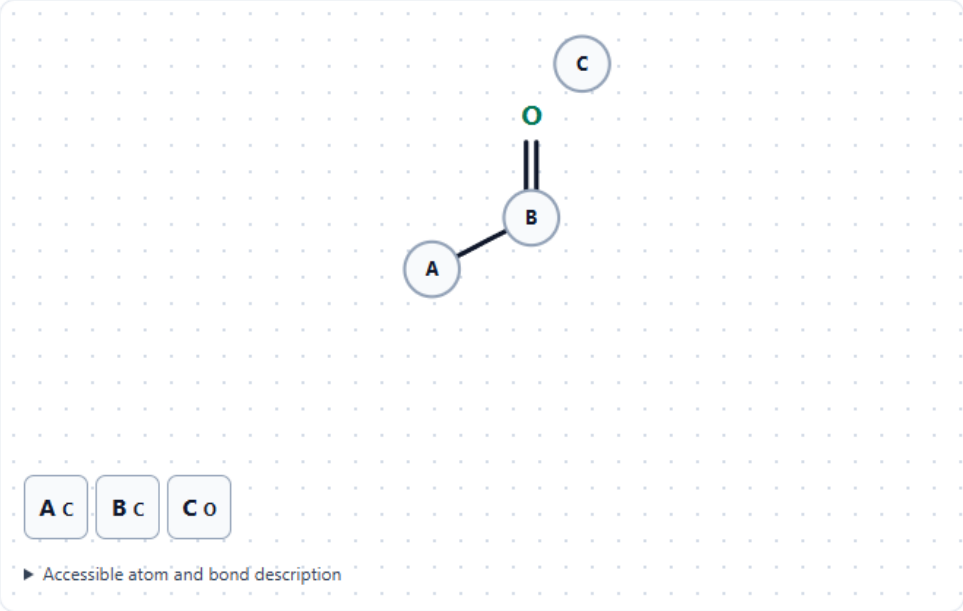
Completion summary: SN2 is bimolecular and stereospecific for geometric inversion.

Screenshot: sn2-rate-complete



carbon-redox | Assign carbon oxidation state

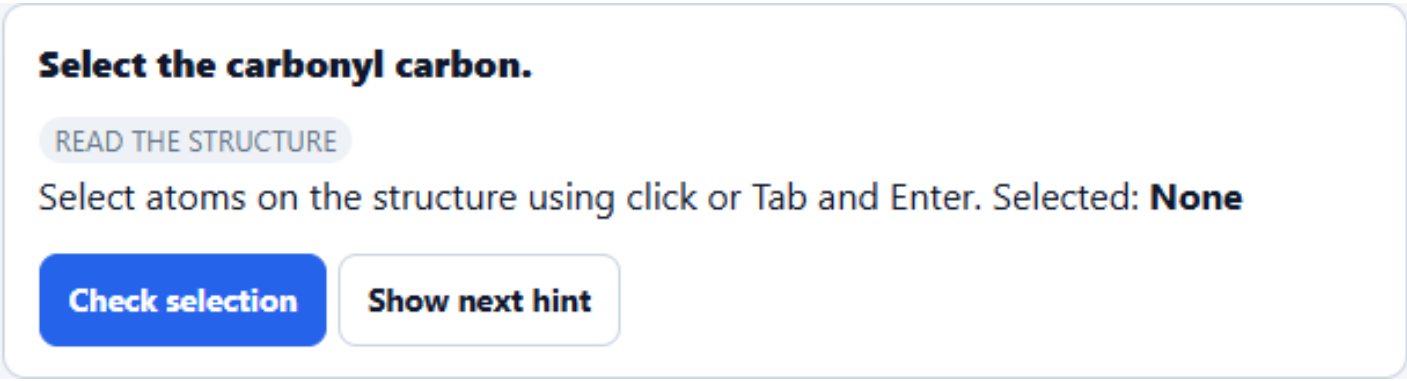
Screenshot: carbon-redox-stage-1-molecule



carbon-redox:stage-1 | Select the carbonyl carbon.

Answer type: atoms

Screenshot: carbon-redox-stage-1-question



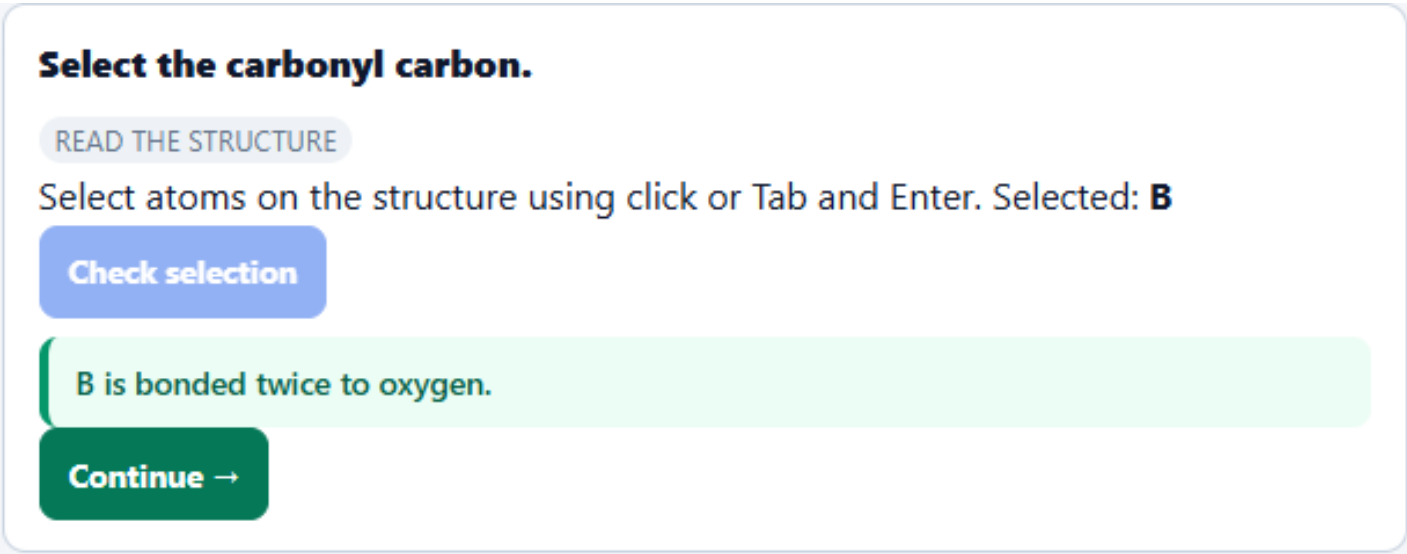
{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1], "sub": [2]}}

Hint 1: Find the C=O bond.

Hint 2: Choose its carbon endpoint.

Explanation: B is bonded twice to oxygen.

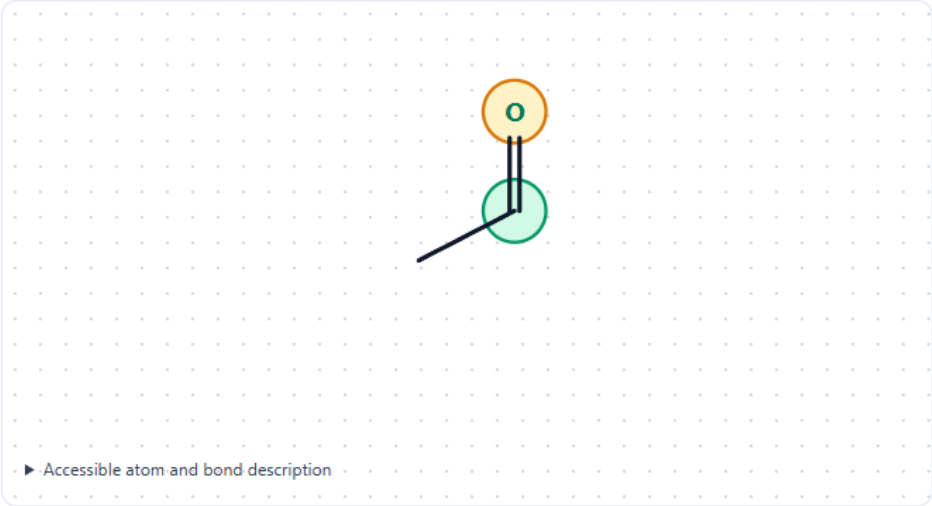
Screenshot: carbon-redox-stage-1-feedback



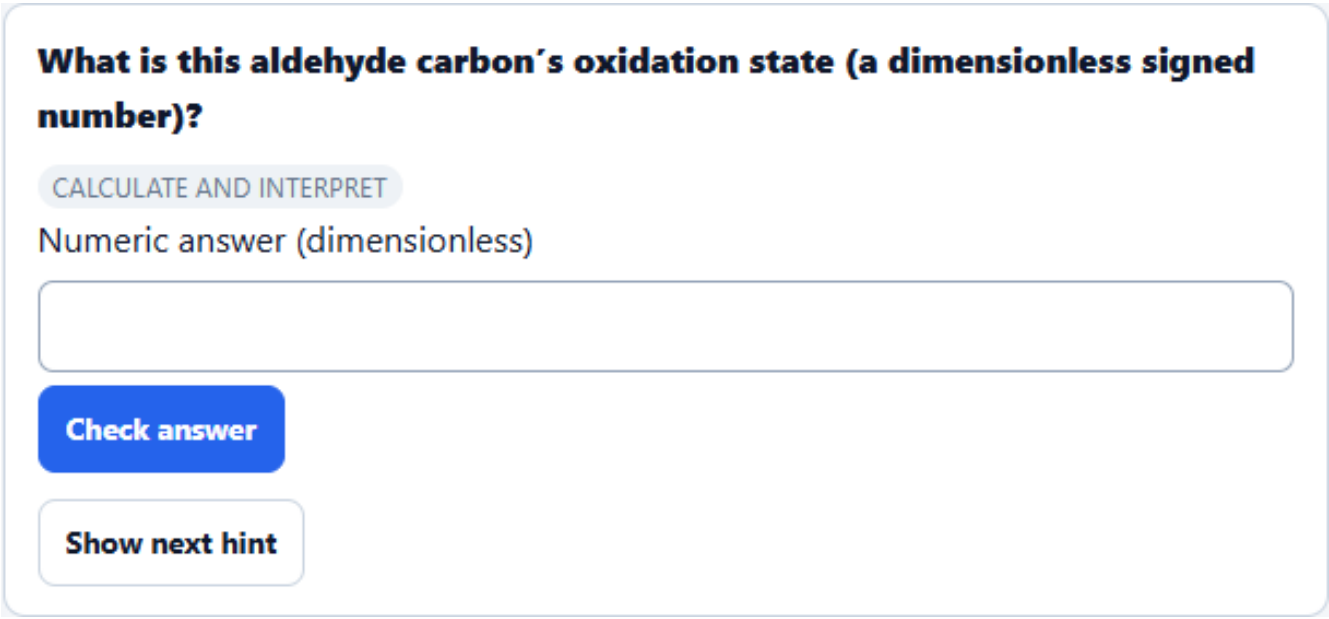
carbon-redox:stage-2 | What is this aldehyde carbon’s oxidation state (a dimensionless signed number)?

Answer type: number | unit: dimensionless

Screenshot: carbon-redox-stage-2-molecule



Screenshot: carbon-redox-stage-2-question



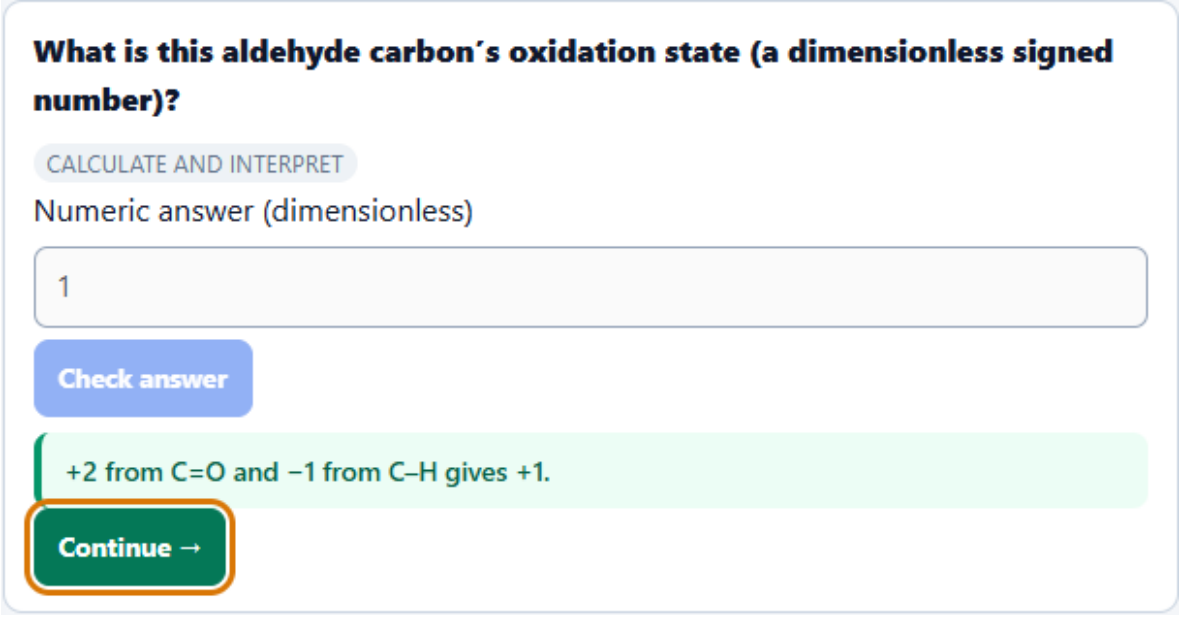
{"authorAnswer": 1, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Assign electrons in C–O to O and in C–H to C.

Hint 2: C–C contributes zero.

Explanation: +2 from C=O and −1 from C–H gives +1.

Screenshot: carbon-redox-stage-2-feedback



carbon-redox:stage-3 | NaBH4 followed by protonation converts it to...

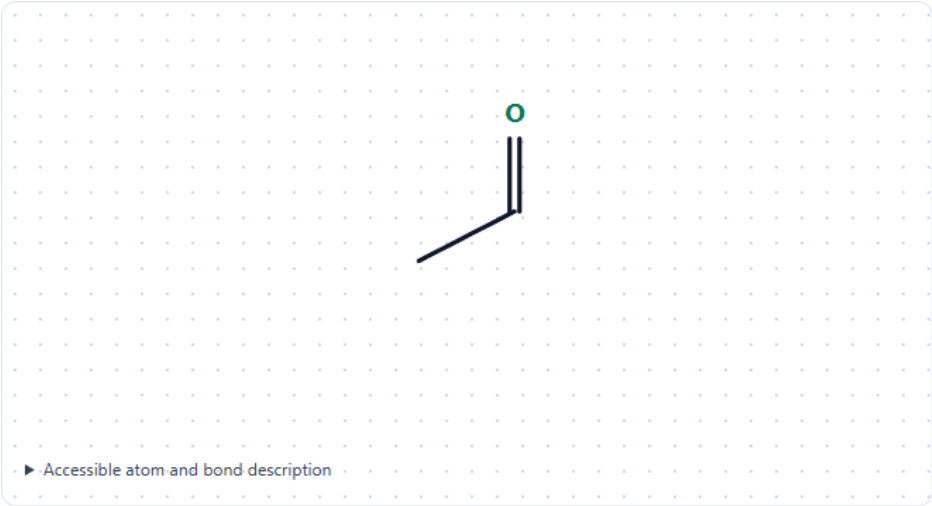
Answer type: choice

Option 0 (A): A primary alcohol

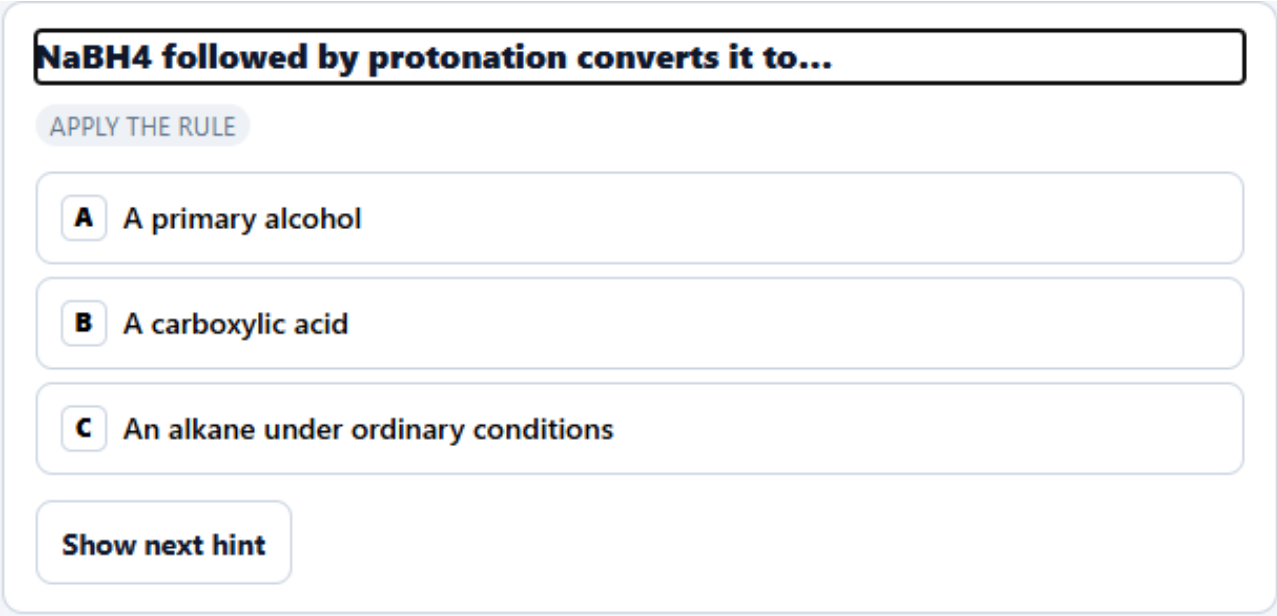
Option 1 (B): A carboxylic acid

Option 2 (C): An alkane under ordinary conditions

Screenshot: carbon-redox-stage-3-molecule



Screenshot: carbon-redox-stage-3-question



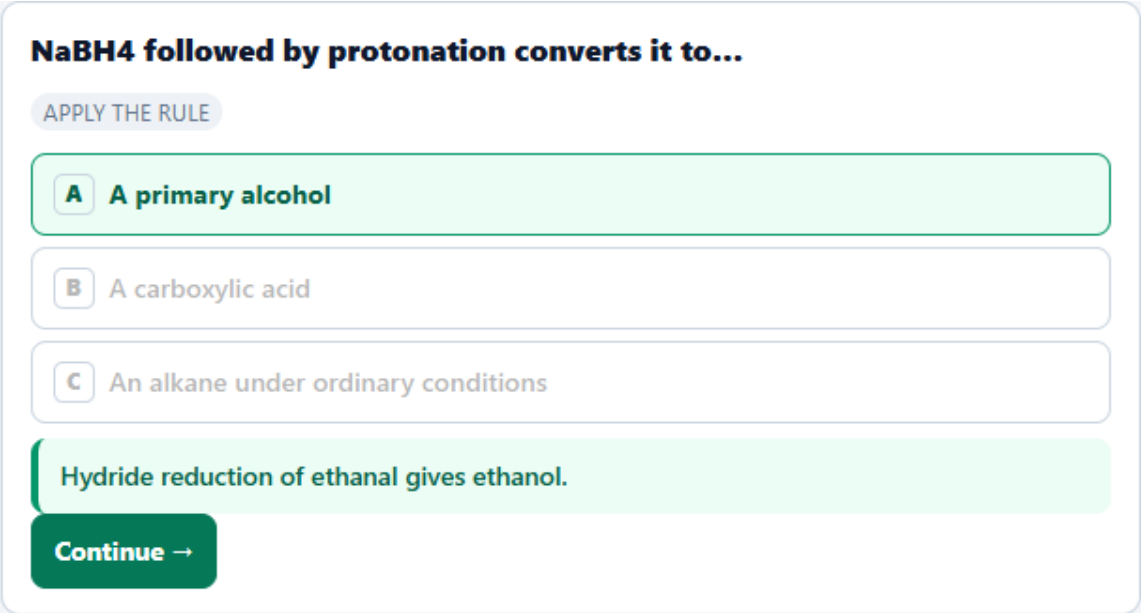
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Hydride adds to C=O.

Hint 2: The oxygen remains in the alcohol product.

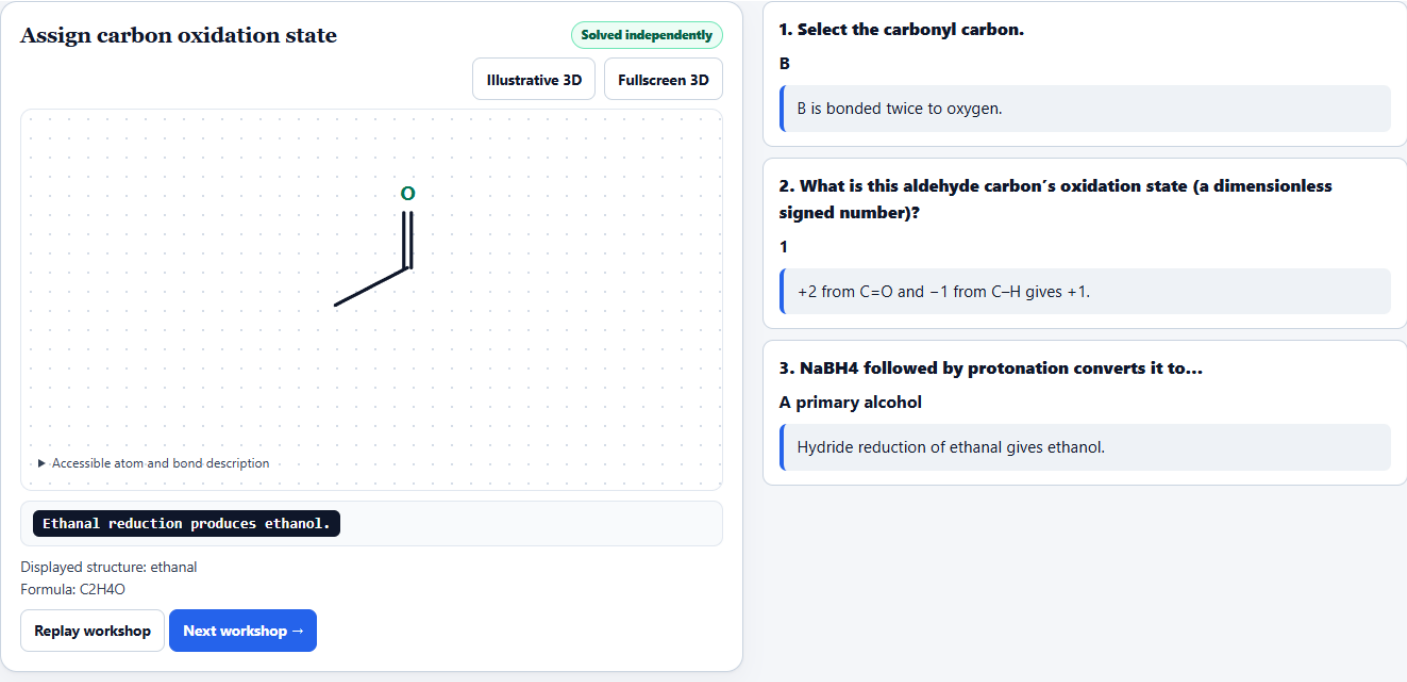
Explanation: Hydride reduction of ethanal gives ethanol.

Screenshot: carbon-redox-stage-3-feedback



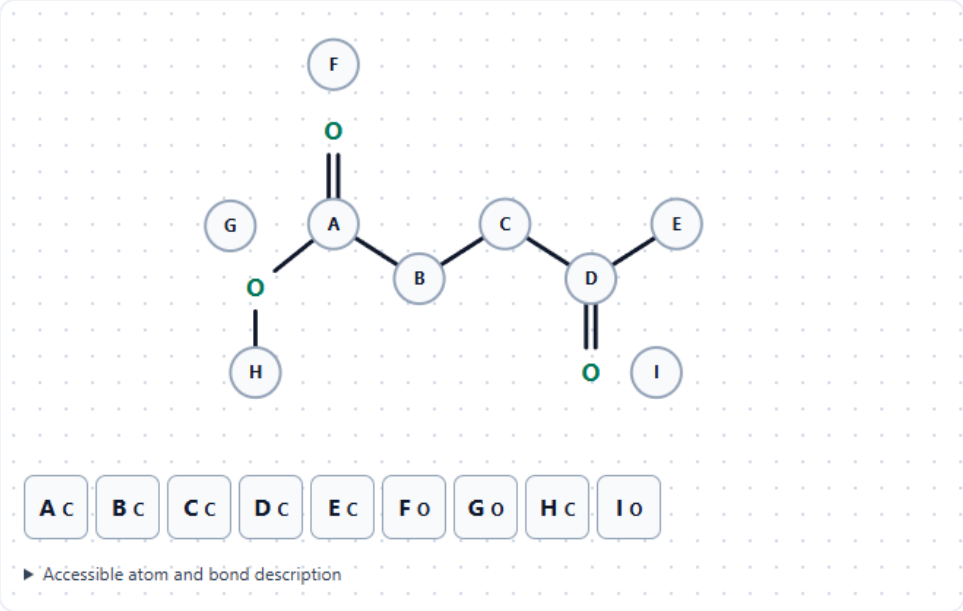
Completion summary: Ethanal reduction produces ethanol.

Screenshot: carbon-redox-complete



selective-reduction | Select a reactive site

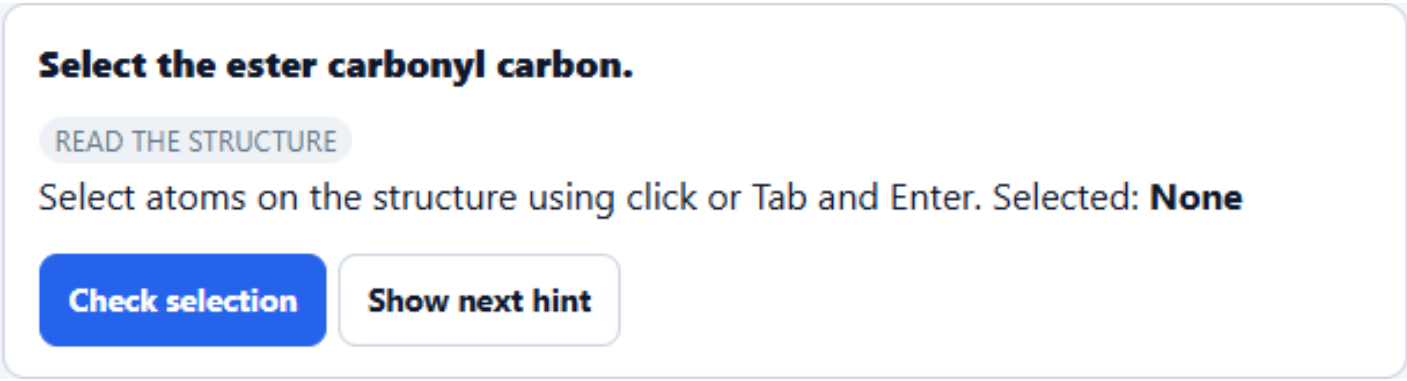
Screenshot: selective-reduction-stage-1-molecule



selective-reduction:stage-1 | Select the ester carbonyl carbon.

Answer type: atoms

Screenshot: selective-reduction-stage-1-question



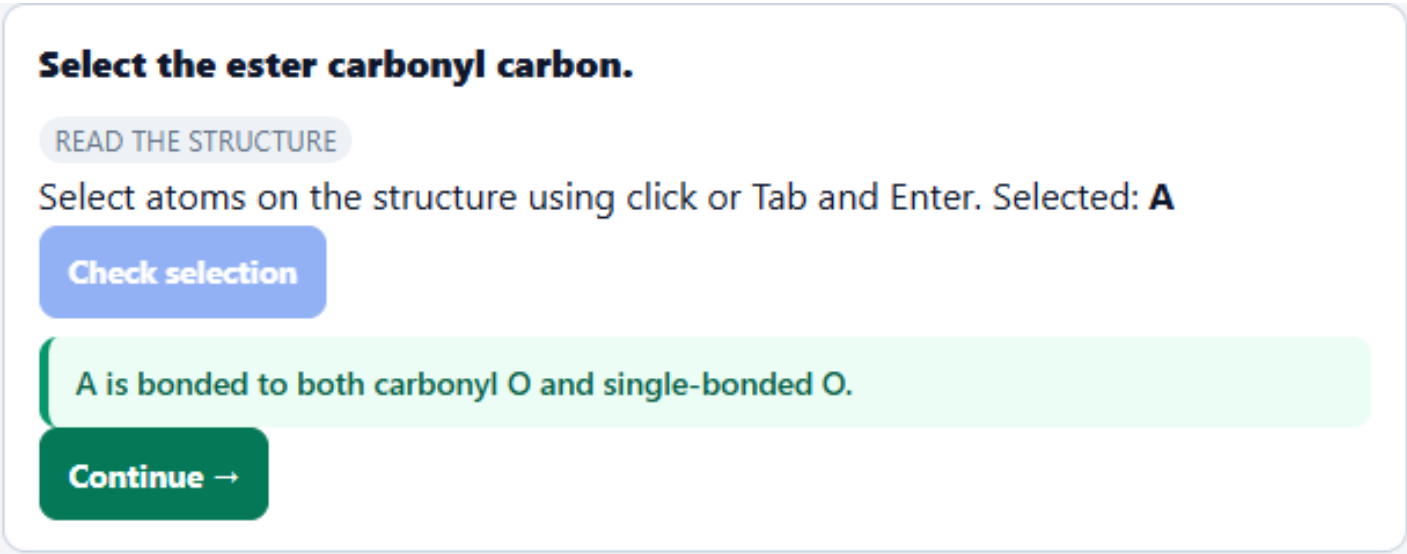
{"authorAnswer": [0], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 5, 6]}}

Hint 1: An ester has C(=O)–O–C.

Hint 2: Find the carbonyl with a second oxygen neighbor.

Explanation: A is bonded to both carbonyl O and single-bonded O.

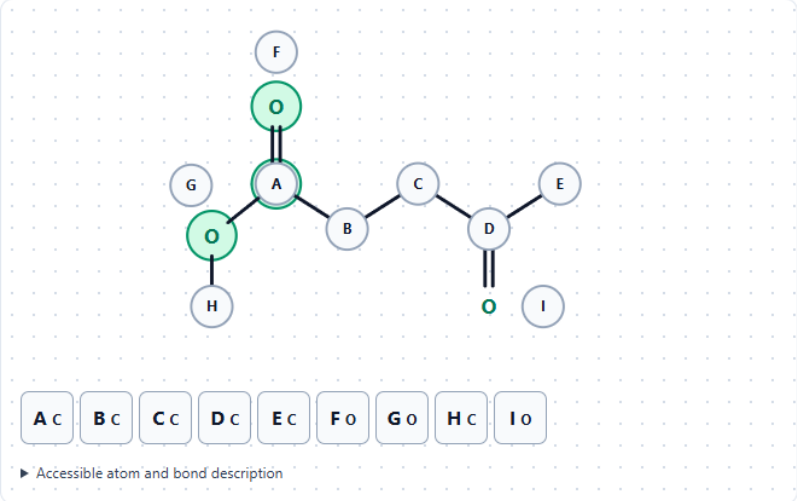
Screenshot: selective-reduction-stage-1-feedback



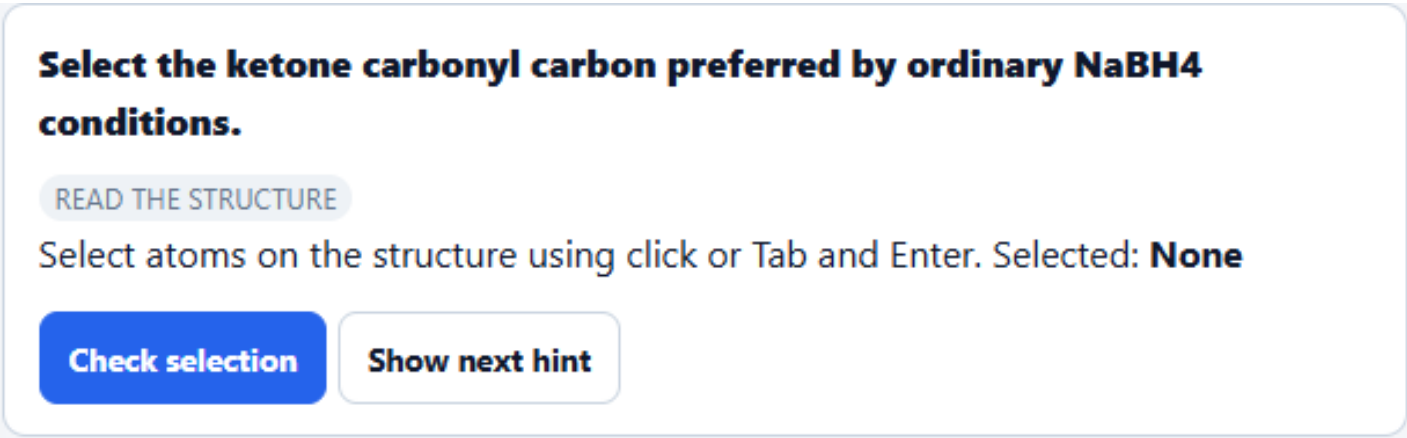
selective-reduction:stage-2 | Select the ketone carbonyl carbon preferred by ordinary NaBH4 conditions.

Answer type: atoms

Screenshot: selective-reduction-stage-2-molecule



Screenshot: selective-reduction-stage-2-question



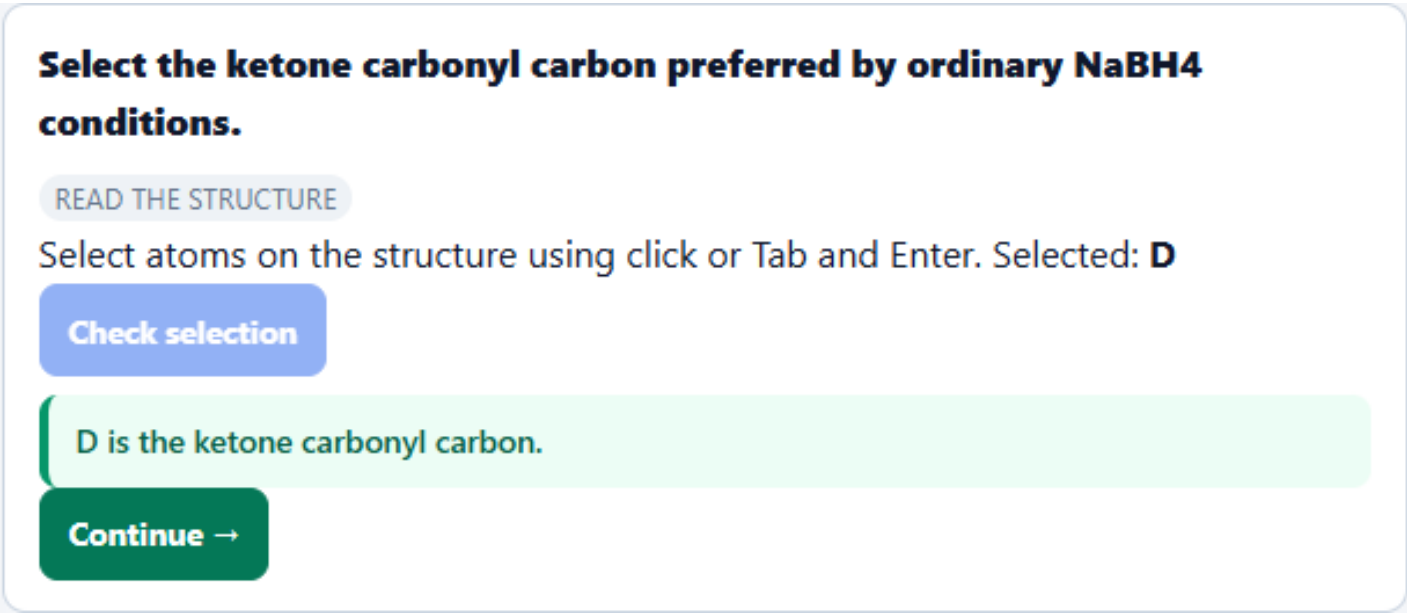
{"authorAnswer": [3], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [3, 8]}}

Hint 1: The ketone has carbon substituents on both sides.

Hint 2: The right-hand carbonyl is more readily reduced here.

Explanation: D is the ketone carbonyl carbon.

Screenshot: selective-reduction-stage-2-feedback



selective-reduction:stage-3 | Why not simply select the highest-priority naming group?

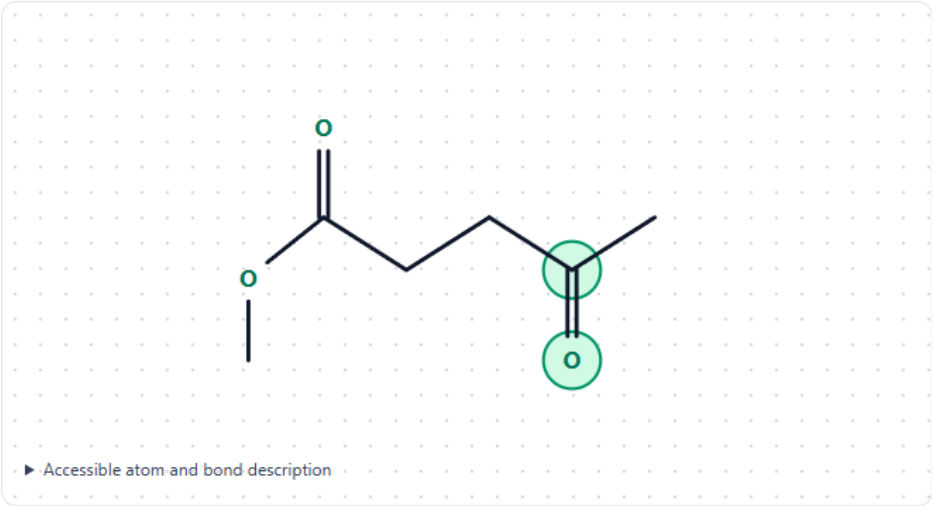
Answer type: choice

Option 0 (A): Naming seniority does not determine kinetic reactivity

Option 1 (B): The ester has no carbonyl

Option 2 (C): All carbonyls react identically

Screenshot: selective-reduction-stage-3-molecule



Screenshot: selective-reduction-stage-3-question

Why not simply select the highest-priority naming group?

APPLY THE RULE

A Naming seniority does not determine kinetic reactivity

B The ester has no carbonyl

C All carbonyls react identically

Show next hint

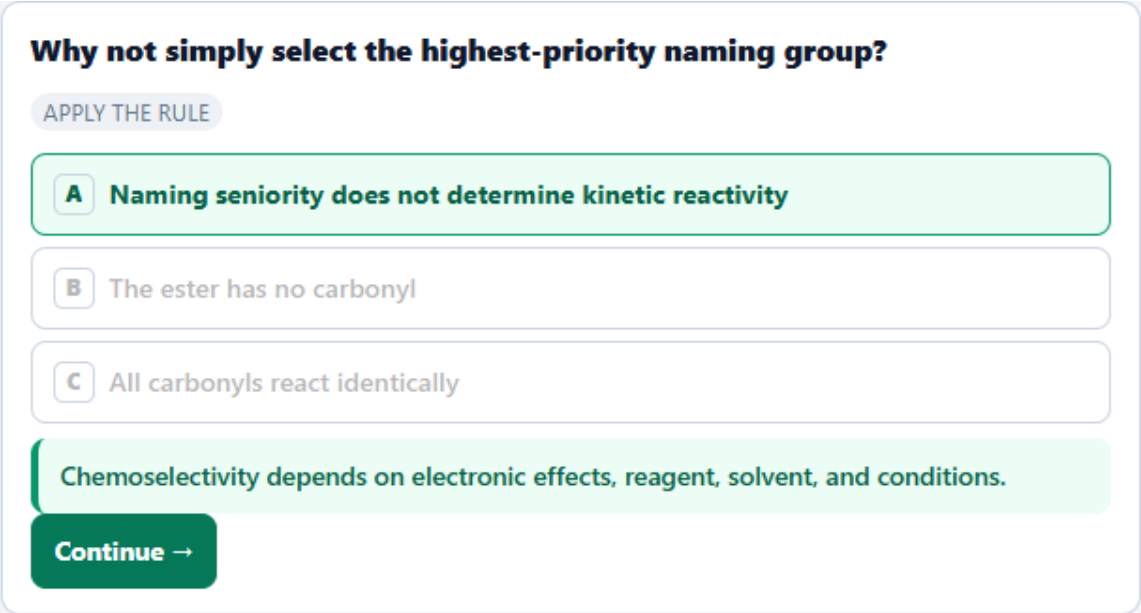
{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Naming and reaction mechanisms solve different problems.

Hint 2: Compare resonance donation and electrophilicity.

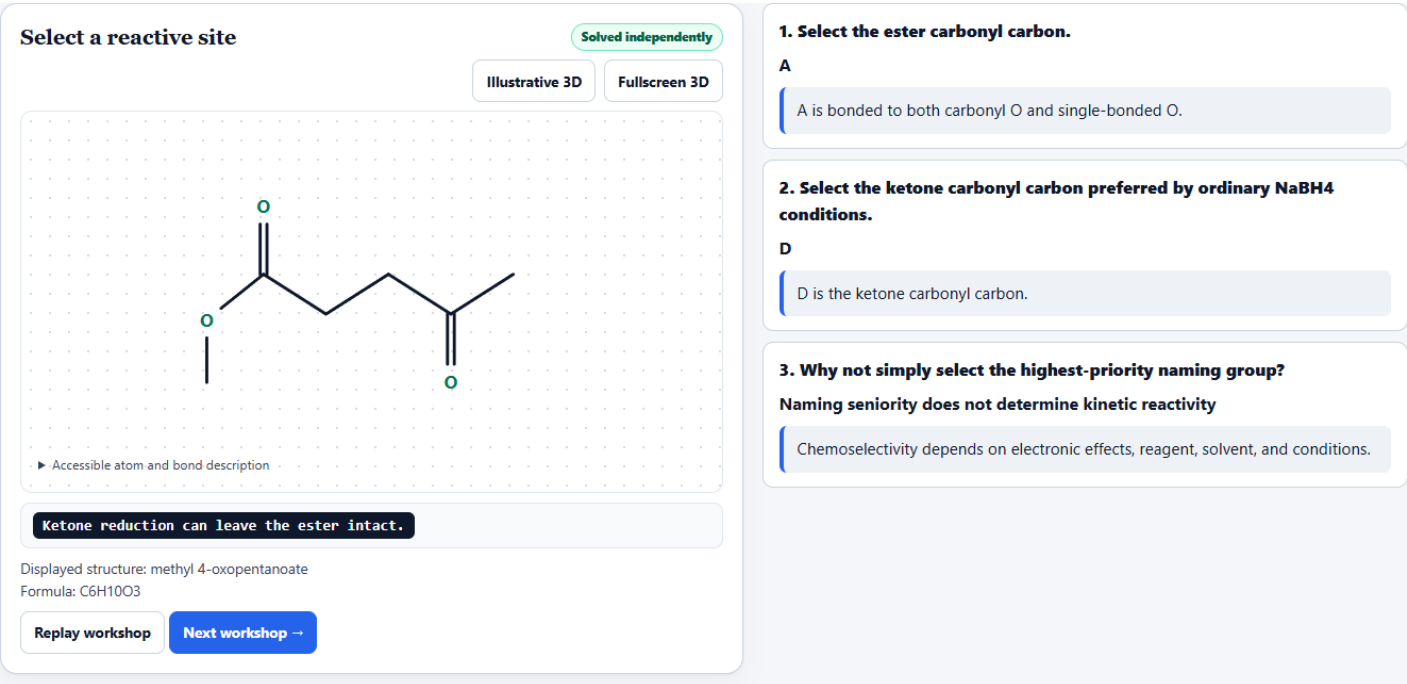
Explanation: Chemoselectivity depends on electronic effects, reagent, solvent, and conditions.

Screenshot: selective-reduction-stage-3-feedback



Completion summary: Ketone reduction can leave the ester intact.

Screenshot: selective-reduction-complete



Molecular repository: author claims to verify

aceticacid

Author claims: ethanoic acid | formula C2H4O2 | canonical composition C2H4O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=108.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

acetate

Author claims: acetate (one resonance contributor) | formula C2H3O2 | canonical composition C2H3O2 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=115.00; charge=-1; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

acetaldehyde

Author claims: ethanal | formula C2H4O | canonical composition C2H4O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=131.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=189.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=189.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2]]

bromoethane

Author claims: bromoethane | formula C2H5Br | canonical composition C2H5Br | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): Br; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

tbutylbromide

Author claims: 2-bromo-2-methylpropane | formula C4H9Br | canonical composition C4H9Br | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=85.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=168.00, y=85.00; charge=0; explicit H override=none

3 (D): C; x=108.00, y=25.00; charge=0; explicit H override=none

4 (E): Br; x=108.00, y=145.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

ethanol

Author claims: ethanol | formula C2H6O | canonical composition C2H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=166.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1]]

acetone

Author claims: propanone | formula C3H6O | canonical composition C3H6O | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=50.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=108.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=166.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=108.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 2]]

ketoester

Author claims: methyl 4-oxopentanoate | formula C6H10O3 | canonical composition C6H10O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=70.00, y=100.00; charge=0; explicit H override=none

1 (B): C; x=125.00, y=135.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=100.00; charge=0; explicit H override=none

3 (D): C; x=235.00, y=135.00; charge=0; explicit H override=none

4 (E): C; x=290.00, y=100.00; charge=0; explicit H override=none

5 (F): O; x=70.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=20.00, y=140.00; charge=0; explicit H override=none

7 (H): C; x=20.00, y=195.00; charge=0; explicit H override=none

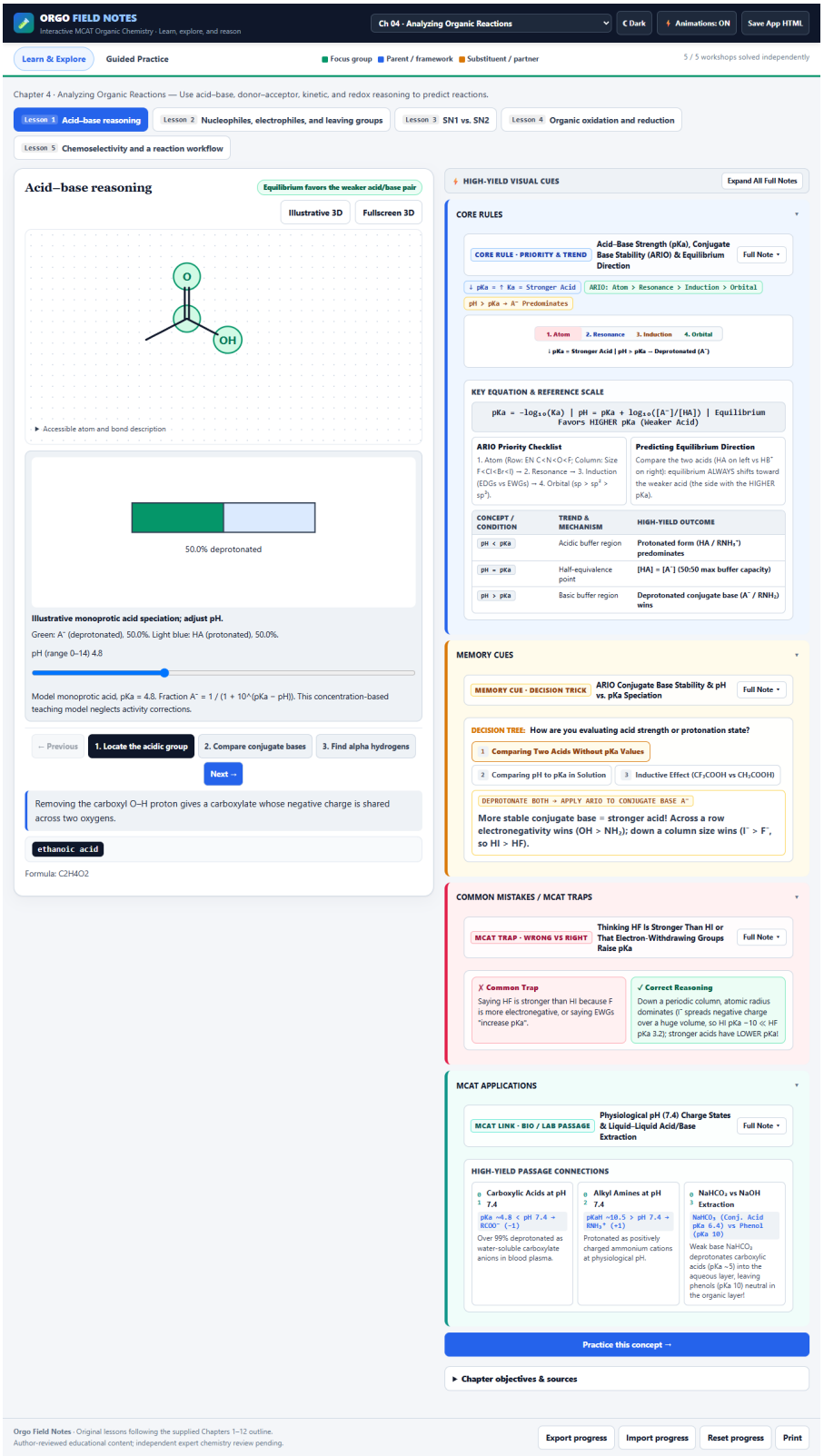
8 (I): O; x=235.00, y=195.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [0, 5, 2], [0, 6, 1], [6, 7, 1], [3, 8, 2]]

Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

Screenshot: layout-desktop-light



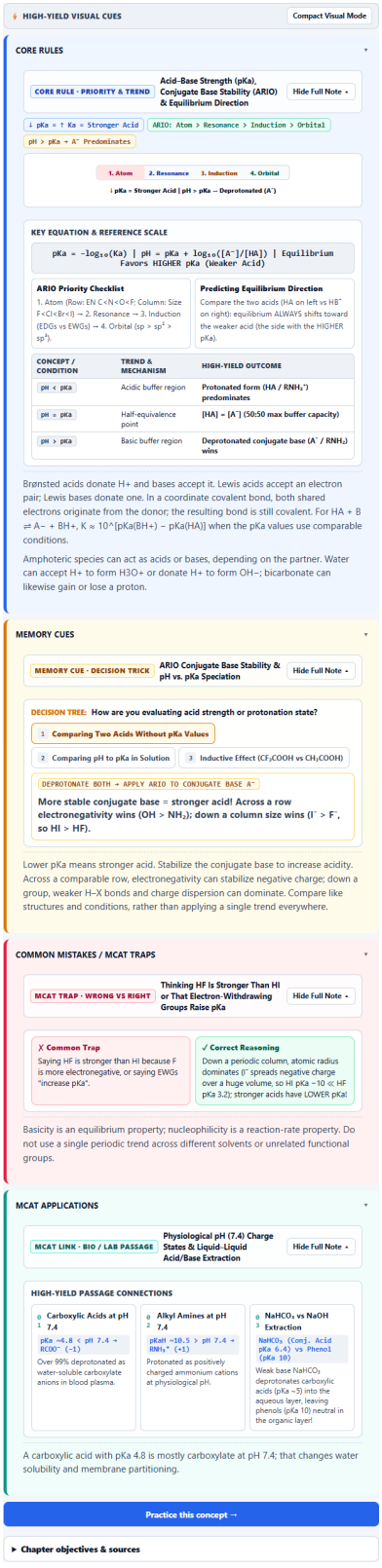
Screenshot: layout-mobile-light



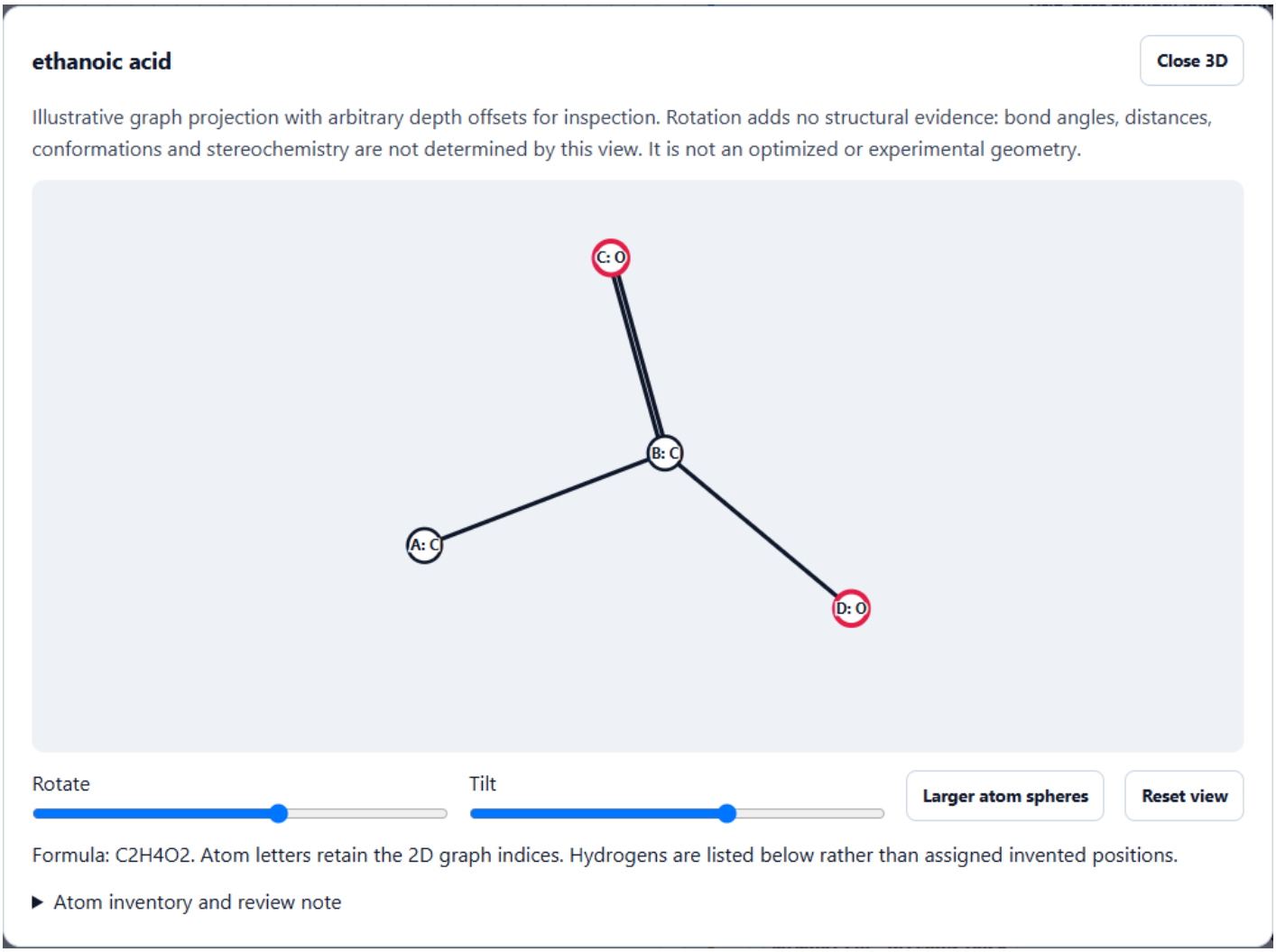
Screenshot: layout-mobile-dark



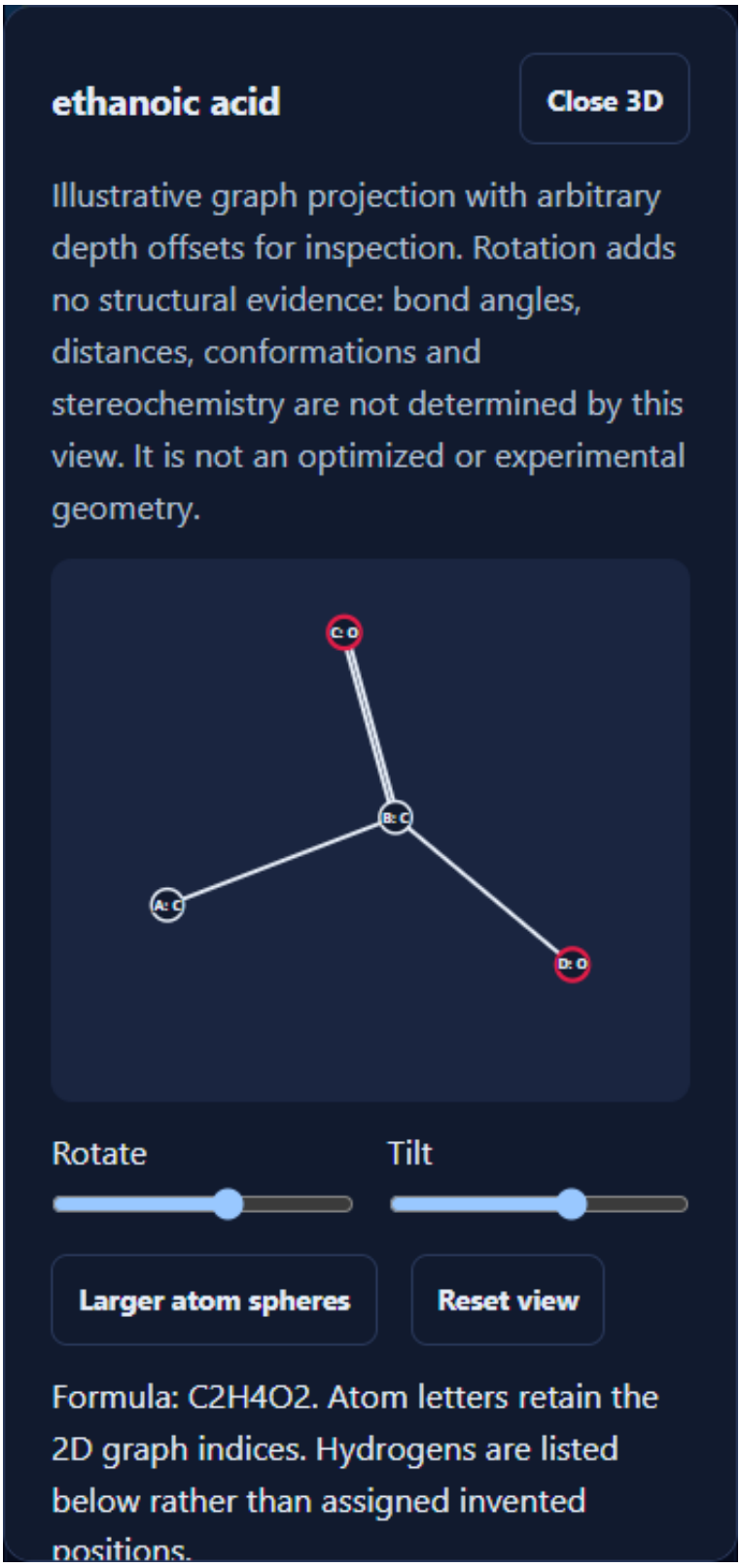
Screenshot: visual-notes-expanded



Screenshot: projection-desktop-light



Screenshot: projection-mobile-dark



Execution evidence

77 screenshots and 18 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.